Linear discriminant functions

Plan

1. Linear functions and linear discriminant functions
2. Different classifiers from different distributional assumptions
   - Intermezzo Normal Distribution and $\mathbf{x}^T\mathbf{W}\mathbf{x}$.
3. Fisher's linear discriminants
   - Intermezzo, Rayleigh quotient
4. Perceptron
   - Controlling the learning rate
5. Logistic Regression

Linear Functions I

A linear discriminant (LD) is a linear combination of the components of $\mathbf{x} = \begin{pmatrix} x_2 \\ x_2 \\ \dots \\ x_d \end{pmatrix}$

given by a *weight vector*, $\mathbf{w} = \begin{pmatrix} w_1 \\ w_2 \\ \dots \\ w_d \end{pmatrix}$,

and a bias parameter, w_0, also known as the *threshold weight*.

Linear Functions II

Then LD can be written as:

$$g(\mathbf{x}) = \mathbf{w}^T\mathbf{x} + w_0$$

or as:

$$g(\mathbf{x}) = \langle \mathbf{w}, \mathbf{x} \rangle + w_0$$

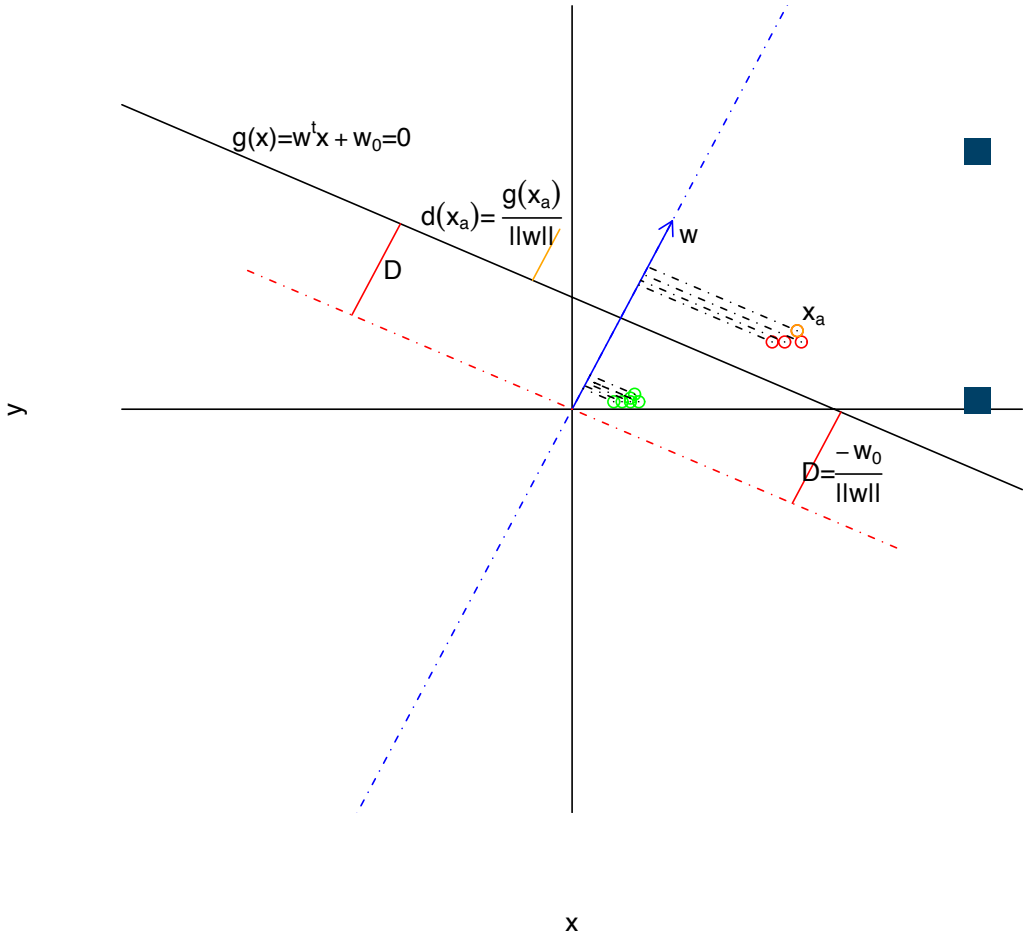
We can also extend the $\mathbf{x}$ vector by a dimension, x_0, fixed to 1, then $\mathbf{x}' := \begin{pmatrix} x_0 \\ \mathbf{x} \end{pmatrix}$ and $\mathbf{w}' := \begin{pmatrix} w_0 \\ \mathbf{w} \end{pmatrix}$ and the LD equation simplifies to:

$$g(\mathbf{x}') = \mathbf{w}'^T\mathbf{x}'$$

Linear Discriminant Functions

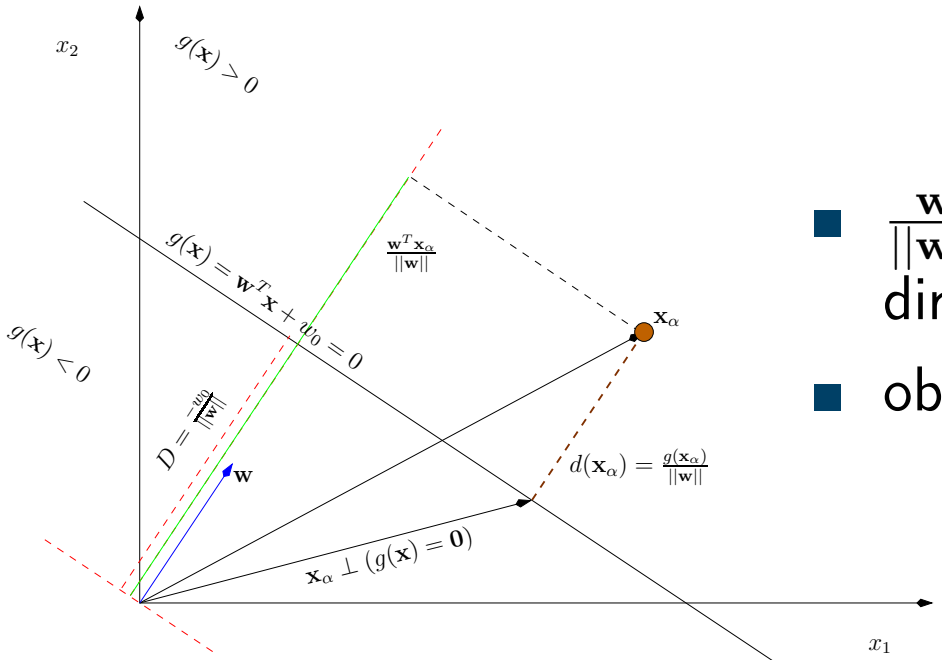
- In the simplest classification scenario we have two classes, ω_1, ω_2, the decision rule of a LD is:
  - If $g(\mathbf{x}) > 0$ then ω_1
  - If $g(\mathbf{x}) < 0$ then ω_2
- $g(\mathbf{x}) = 0$ defines the decision surface that separates the two classes. When $g(\mathbf{x})$ is linear the decision surface, i.e. $g(\mathbf{x})$, is a *hyperplane*.
- $\mathbf{w}$ is the vector that is normal to the hyperplane $g(\mathbf{x}) = 0$

Some notions from geometry I



- $d(\mathbf{x}_a) = \frac{g(\mathbf{x}_\alpha)}{||\mathbf{w}||}$ is the *signed* distance of $\mathbf{x}_\alpha$ from the $g(\mathbf{x}) = 0$ hyperplane.
- $D = -d(\mathbf{x}_a = \mathbf{0}) = -\frac{w_0}{||\mathbf{w}||}$ is the *signed* distance of the $g(\mathbf{x}) = 0$ hyperplane from the origin of the axes.

Distance, $d(\mathbf{x}_\alpha)$, of a point $\mathbf{x}_\alpha$ from the hyperplane $g(\mathbf{x}) = 0$, $d(\mathbf{x}_\alpha) = \frac{g(\mathbf{x}_\alpha)}{||\mathbf{w}||}$



- Lets denote the projection point of $\mathbf{x}_\alpha$ on the decision surface $g(\mathbf{x}) = 0$ by $\mathbf{x}_{\alpha_\perp}$, i.e.

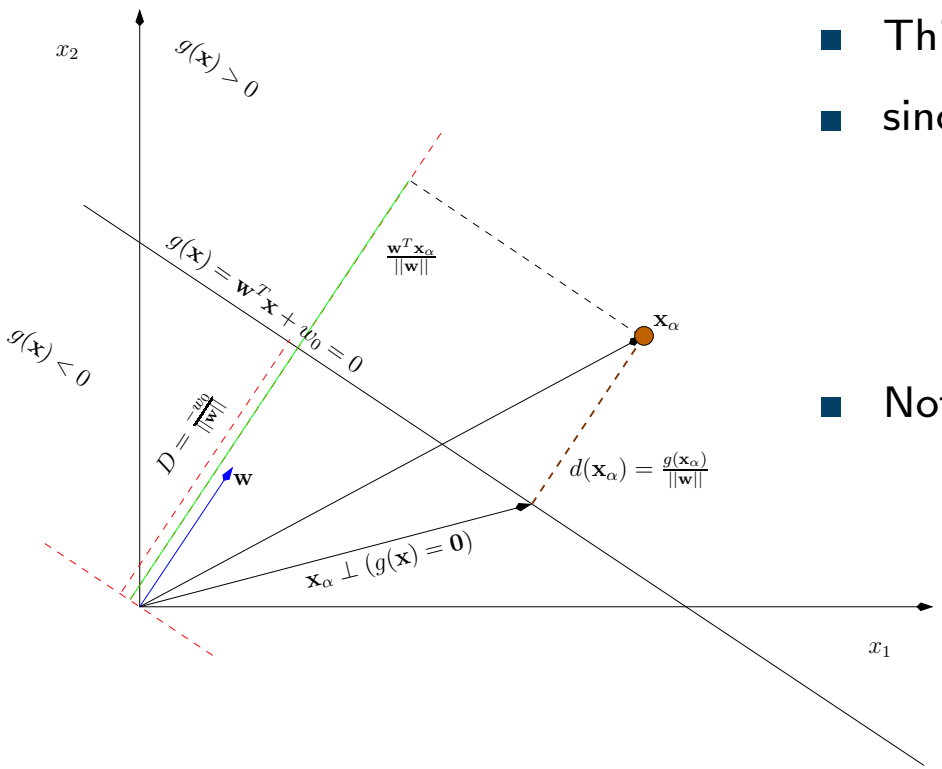
$$\mathbf{x}_{\alpha_\perp} := \mathbf{x}_\alpha \perp (g(\mathbf{x}) = 0)$$

- $\frac{\mathbf{w}}{||\mathbf{w}||}$ is the unit length vector on the direction of $\mathbf{w}$
- obviously

$$\mathbf{x}_\alpha = \mathbf{x}_{\alpha_\perp} + d(\mathbf{x}_\alpha)\frac{\mathbf{w}}{||\mathbf{w}||} \tag{1}$$

- By multiplying eq 1 by $\mathbf{w}^T$ and adding w_0 we can show that $d(\mathbf{x}_\alpha) = \frac{g(\mathbf{x}_\alpha)}{||\mathbf{w}||}$

Distance, D, of the hyperplane $g(\mathbf{x}) = 0$ from the origin of axis, $D = -d(\mathbf{x}_\alpha = 0) = -\frac{w_0}{||\mathbf{w}||}$



- For any $\mathbf{x} \in g(\mathbf{x}) = 0$ its projection on $\frac{\mathbf{w}}{||\mathbf{w}||}$ gives the signed distance, D, of the hyperplane from the origin.
- This projection is given by $\frac{\mathbf{w}^T}{||\mathbf{w}||}\mathbf{x}$,
- since $\mathbf{x} \in g(\mathbf{x}) = 0$ we have

$$D = \frac{\mathbf{w}^T}{||\mathbf{w}||}\mathbf{x} = -\frac{w_0}{||\mathbf{w}||} = -d(\mathbf{x}_\alpha = 0)$$

- Note that:

$$\begin{aligned} d(\mathbf{x}_\alpha) &= \frac{g(\mathbf{x}_\alpha)}{||\mathbf{w}||} \\ &= \frac{\mathbf{w}^T\mathbf{x}_\alpha}{||\mathbf{w}||} + \frac{w_0}{||\mathbf{w}||} \\ &= \frac{\mathbf{w}^T\mathbf{x}_\alpha}{||\mathbf{w}||} - D \end{aligned}$$

Summarizing

For the hyperplane $g(\mathbf{x}) = \mathbf{w}^T\mathbf{x} + w_0 = 0$ we have:

- $\mathbf{w}$ is the vector normal to the hyperplane and defines its direction
- w_0 defines the position of the hyperplane with respect to the origin of the axes, remember that:
  - $D = -\frac{w_0}{||\mathbf{w}||}$ gives the signed distance of the hyperplane from the origin of the axes.
  - $d(\mathbf{x}_\alpha = \mathbf{0}) = -D$ gives the signed distance of the origin of the axes from the hyperplane.
- For any point $\mathbf{x_a}$,
  - sign$(g(\mathbf{x_a}))$ tells us on which side of the hyperplane $g(\mathbf{x}) = 0$ is $\mathbf{x}_a$.
  - $d(\mathbf{x_a}) = \frac{g(\mathbf{x_a})}{||\mathbf{w}||}$, gives its signed distance from the hyperplane $g(\mathbf{x}) = 0$.

How to compute $g(\mathbf{x}) = 0$

There are many different ways to compute the $\mathbf{w}$, and w_0, needed for the definition of $g(\mathbf{x})$. Just some of them:

- Deriving linear discriminants, and more, using Bayes theorem and assumptions about the distribution governing the data.
- Fisher's linear discriminants
- Perceptron
- Logistic Regression
- Support Vector Machines

LD (and not only) using distributional assumptions I

- Starting from the Bayes theorem we can construct:
  linear or *quadratic*
  discriminant functions
- assuming that instances of each class, c_i, follow a multivariate normal distribution $P(\mathbf{x}|c_i) \sim N(\boldsymbol{\mu}_i, \boldsymbol{\Sigma}_i)$.
- From Bayes theorem we have:

$$
\begin{aligned}
\arg\max_{c_i}(P(c_i|\mathbf{x})) &= \arg\max_{c_i} \frac{P(\mathbf{x}|c_i)P(c_i)}{P(\mathbf{x})} \\
&= \arg\max_{c_i} \frac{P(\mathbf{x}|c_i)P(c_i)}{\sum_j P(\mathbf{x}|c_j)P(c_j)} \\
&= \arg\max_{c_i}(P(\mathbf{x}|c_i)P(c_i)) \\
&= \arg\max_{c_i}(\ln(P(\mathbf{x}|c_i)) + \ln(P(c_i)))
\end{aligned}
$$

LD (and not only) using distributional assumptions II

- Since $P(\mathbf{x}|c_i) \sim N(\boldsymbol{\mu}_i, \boldsymbol{\Sigma}_i) = \frac{1}{(2\pi)^{d/2}|\boldsymbol{\Sigma}_i|^{1/2}} e^{-\frac{1}{2}(\mathbf{x}-\boldsymbol{\mu_i})^T \boldsymbol{\Sigma}_i^{-1}(\mathbf{x}-\boldsymbol{\mu_i})}$
- Then we have the decision rule:

$$\begin{aligned}\arg\max_{c_i}(P(c_i|\mathbf{x})) &= \arg\max_{c_i}[\ln(\frac{1}{(2\pi)^{d/2}|\boldsymbol{\Sigma}_i|^{1/2}}) \\ &\quad -\frac{1}{2}(\mathbf{x}-\boldsymbol{\mu}_i)^T\boldsymbol{\Sigma}_i^{-1}(\mathbf{x}-\boldsymbol{\mu_i}) \\ &\quad +\ln(P(c_i))]\end{aligned}$$

- Note the presence of the *Mahalanobis* distance, $(\mathbf{x}-\boldsymbol{\mu_i})^T\boldsymbol{\Sigma}_i^{-1}(\mathbf{x}-\boldsymbol{\mu}_i)$, from the class centroid $\boldsymbol{\mu}_i$.
- "All" that is left is estimating $\boldsymbol{\Sigma}_i$s, $\boldsymbol{\mu}_i$s

Learning problem formulation, LD using distributional assumptions

- We have made a number of distributional assumptions
- Now we need to:
  - Define an optimization problem
    - We should define an objective function (based on our distributional assumptions)
    - which we minimize or maximize depending on the function we choose with respect to its parameters.
  - Solve it, i.e. find the parameters that optimize it.
- This is the general approach we will follow *all* the time.

Objective function I, Normal Linear Discriminants (NLD)

- Concretely
  - We have specified a parametric functional form for the class conditional densities $P(\mathbf{x}|c_i)$ (normal distributions)
  - We will use as objective function the **likelihood** function
  - which we will maximize with respect to its parameters (here $\boldsymbol{\mu_i}, \boldsymbol{\Sigma_i}, \pi$)
  - over a training set comprised of N $\mathbf{x}_i$ instances and their class labels y_i
- So we are following a *maximum likelihood* approach.

Objective function II, Likelihood function, (NLD)

- Since we will use the maximum likelihood estimation...
- we need to write down the likelihood function of our training set.
- Any ideas????

Objective function II, Likelihood function, (NLD)

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$$p(\mathbf{X}, \mathbf{y}|\boldsymbol{\theta}) = \prod_{i=1}^{N} p(\mathbf{x}_i, y_i|\boldsymbol{\theta})$$

Objective function III, Likelihood function, the $p(\mathbf{x}_i, y_i)$ terms, NLD

- what do we do with the $p(\mathbf{x}_i, y_i)$? How do we compute them?

Objective function III, Likelihood function, the $p(\mathbf{x}_i, y_i)$ terms, NLD

- what do we do with the $p(\mathbf{x}_i, y_i)$? How do we compute them?
  - Assume we have two classes C_1 and C_2
  - with class labels $y = 1$ and $y = 0$ respectively,
  - and $p(C_1) = \pi$.
- For a data point x_i of class C_1 we have $y_i = 1$ and:

$$p(\mathbf{x}_i, y_i|\boldsymbol{\theta}) = p(\mathbf{x}_i, C_1|\boldsymbol{\theta}) =$$

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$$p(\mathbf{x}_i, y_i|\boldsymbol{\theta}) = p(\mathbf{x}_i, C_1|\boldsymbol{\theta}) = p(C_1)p(\mathbf{x}_i|C_1) = \pi N(\mathbf{x}_i|\boldsymbol{\mu_1}, \boldsymbol{\Sigma_1})$$

- for a data point x_i of class C_2 we have $y_i = 0$ and:

$$p(\mathbf{x}_i, y_i|\boldsymbol{\theta}) = p(\mathbf{x}_i, C_2|\boldsymbol{\theta}) =$$

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- for a data point x_i of class C_2 we have $y_i = 0$ and:

$$p(\mathbf{x}_i, y_i|\boldsymbol{\theta}) = p(\mathbf{x}_i, C_2|\boldsymbol{\theta}) = p(C_2)p(\mathbf{x}_i|C_2) = (1-\pi)N(\mathbf{x}_i|\boldsymbol{\mu_2}, \boldsymbol{\Sigma_2})$$

- $\boldsymbol{\theta} = \{\pi, \boldsymbol{\mu}_1, \boldsymbol{\Sigma}_1, \boldsymbol{\mu}_2, \boldsymbol{\Sigma}_2\}$ the parameters of the likelihood function

Objective function III, Likelihood function, the $p(\mathbf{x}_i, y_i)$ terms, NLD

- How do I write a general function form of $\prod_{i=1}^{N} p(\mathbf{x}_i, y_i)$
- i.e. how do I state that I should choose over

$$p(\mathbf{x}_i, y_i|\boldsymbol{\theta}) = p(\mathbf{x}_i, C_1|\boldsymbol{\theta}) = p(C_1)p(\mathbf{x}_i|C_1) = \pi N(\mathbf{x}_i|\boldsymbol{\mu_1}, \boldsymbol{\Sigma_1})$$

or

$$p(\mathbf{x}_i, y_i|\boldsymbol{\theta}) = p(\mathbf{x}_i, C_2|\boldsymbol{\theta}) = p(C_2)p(\mathbf{x}_i|C_2) = (1-\pi)N(\mathbf{x}_i|\boldsymbol{\mu_2}, \boldsymbol{\Sigma_2})$$

for the ith term of the product????

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for the ith term of the product???? Solution:

$$[\pi N(\mathbf{x}_i|\boldsymbol{\mu}_1, \boldsymbol{\Sigma_1}]^{y_i}[(1-\pi)N(\mathbf{x}_i|\boldsymbol{\mu}_2, \boldsymbol{\Sigma_2}]^{1-y_i}$$

Objective function III, NLD

- then the likelihood function is:

$$p(\mathbf{X}, \mathbf{y}|\pi, \boldsymbol{\mu_1}, \boldsymbol{\mu_2}, \boldsymbol{\Sigma}_1, \boldsymbol{\Sigma}_2) = \prod_{i=1}^{N} [\pi N(\mathbf{x}_i|\boldsymbol{\mu}_1, \boldsymbol{\Sigma_1}]^{y_i} [(1-\pi) N(\mathbf{x}_i|\boldsymbol{\mu}_2, \boldsymbol{\Sigma_2}]^{1-y_i}$$

  where $\mathbf{y} = (y_1, y_2, \ldots, y_n)^\mathsf{T}$ the class labels of the training instances

- and the optimization problem is:

$$\arg \max_{\pi, \boldsymbol{\mu_1}, \boldsymbol{\mu_2}, \boldsymbol{\Sigma}_1, \boldsymbol{\Sigma}_2} p(\mathbf{X}, \mathbf{y}|\pi, \boldsymbol{\mu_1}, \boldsymbol{\mu_2}, \boldsymbol{\Sigma}_1, \boldsymbol{\Sigma}_2)$$

- How will we optimize it???

Objective function III, NLD

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- How will we optimize it??? by taking partial derivatives and setting them to zero.

Optimizing the likelihood I

- It is much more convinient to maximize the $\log$ of the likelihood

$$\begin{aligned}\ln p(\mathbf{X}, \mathbf{y}|\pi, \boldsymbol{\mu_1}, \boldsymbol{\mu_2}, \boldsymbol{\Sigma}_1, \boldsymbol{\Sigma}_2) &= \sum_{i=1}^{N} y_i \ln[\pi N(\mathbf{x}_i|\boldsymbol{\mu}_1, \boldsymbol{\Sigma_1})] \\ &+ \sum_{i=1}^{N} (1-y_i) \ln[(1-\pi) N(\mathbf{x}_i|\boldsymbol{\mu}_2, \boldsymbol{\Sigma_2})]\end{aligned}$$

- We will first look to maximize with respect to π,
- Extracting the terms of the likelihood that depend on π the likelihood function will be:

$$\sum_{i=1}^{N} [y_i \ln \pi + (1-y_i) \ln(1-\pi)]$$

Optimizing the likelihood II

- Setting the derivative with respect to π to 0 and rearranging we get

$$\pi = \sum_{i=1}^{N} y_i = N_1/N$$

where N_1 is the number of instances of C_1 class, i.e. just the class prior.

- Continuing with $\boldsymbol{\mu}_1$ we select from the log likelihood only the terms that depend on $\boldsymbol{\mu}_1$:

$$\sum_{i=1}^{N} y_i \ln N(\mathbf{x}_i|\boldsymbol{\mu}_1, \Sigma_1) = -1/2 \sum_{i=1}^{N} y_i (\mathbf{x}_i - \boldsymbol{\mu}_1)^T \boldsymbol{\Sigma}_1^{-1} (\mathbf{x}_i - \boldsymbol{\mu}_1) + const$$

Optimizing the likelihood III

- Getting the derivative with respect to $\boldsymbol{\mu}_1$ and setting to zero we get:

$$\boldsymbol{\mu}_1 = 1/N_1 \sum_{i=1}^{N} y_i \mathbf{x}_i$$

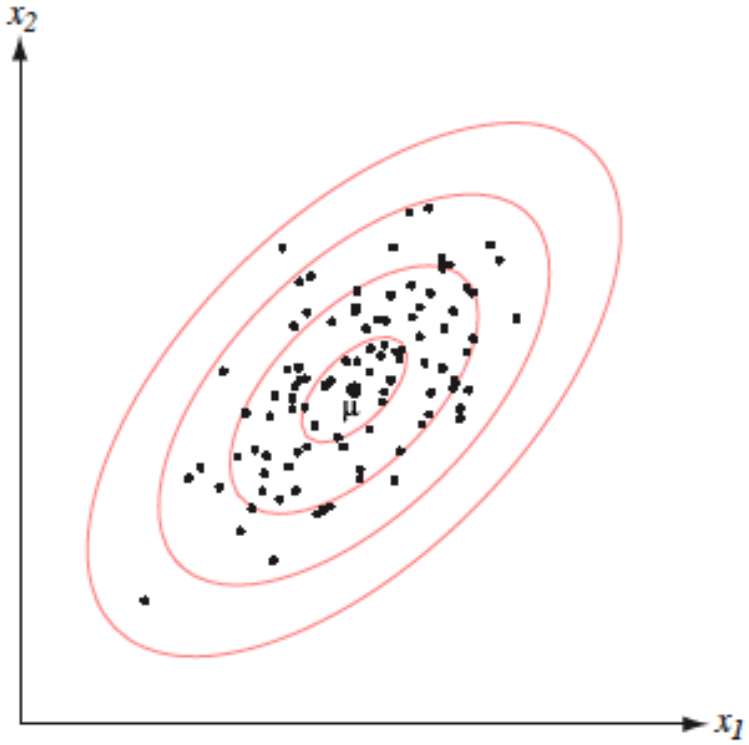
  i.e. just the mean of the C_1 class.
- in the same way for μ_2 we get

$$\boldsymbol{\mu}_2 = 1/(N - N_1) \sum_{i=1}^{N} (1 - y_i) \mathbf{x}_i$$

- In a similar manner we also compute $\boldsymbol{\Sigma}_1, \boldsymbol{\Sigma}_2$
- Thus the Maximum Likelihood estimates are the class priors, the class conditional means, and the class conditional covariance matrices.

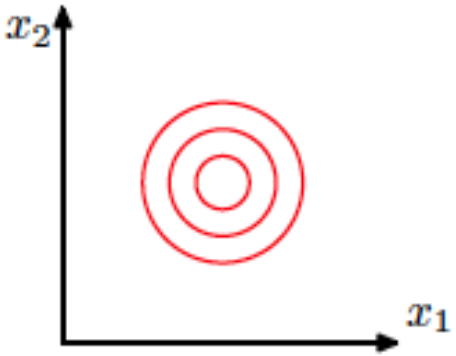
Some notes on the Multivariate Normal Distribution

- $P(\mathbf{x}) \sim N(\boldsymbol{\mu}, \boldsymbol{\Sigma}) = \frac{1}{(2\pi)^{d/2}|\boldsymbol{\Sigma}|^{1/2}} e^{-\frac{1}{2}(\mathbf{x}-\boldsymbol{\mu})^\mathsf{T}\boldsymbol{\Sigma}^{-1}(\mathbf{x}-\boldsymbol{\mu})}$
- $\boldsymbol{\mu} = (\mu_1, \mu_2, ..., \mu_d)^T$, the d-dimensional mean vector
- $\boldsymbol{\Sigma}$ the $d \times d$ covariance matrix
- the normal distribution is completely specified by $d + d(d+1)/2$ parameters
- the position of the data set is determined by the $\boldsymbol{\mu}$
- the shape of the data set is determined by the $\boldsymbol{\Sigma}$
- points of constant density, i.e. $P(\mathbf{x}) = c$, are hyperellipsoids for which the squared *Mahalanobis* distance $(\mathbf{x} - \boldsymbol{\mu})^\mathsf{T}\boldsymbol{\Sigma}^{-1}(\mathbf{x} - \boldsymbol{\mu})$ is constant.

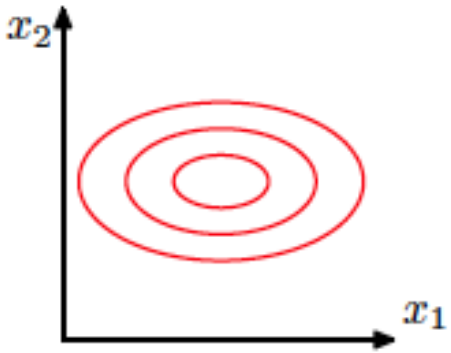


More on the shape of the Multivariate Normal Distribution

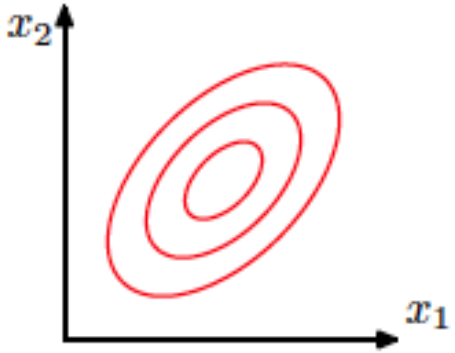
- Different scenarios of the covariance matrix $\mathbf{\Sigma}$.
- Contours represent points of fixed distance from the center $\mathbf{c}$



$\mathbf{\Sigma} = \sigma\mathbf{I}$
contours are circles
circles



$\mathbf{\Sigma}$ is diagonal
contours are
axis-aligned ellipsoids



$\mathbf{\Sigma}$ of general form
contours are
of general form

A bit more on the Multivariate Normal Distribution, MND I

- Remember that the

$$-\frac{1}{2}(\mathbf{x}-\boldsymbol{\mu})^{\mathsf{T}}\boldsymbol{\Sigma}^{-1}(\mathbf{x}-\boldsymbol{\mu}) \tag{2}$$

  Mahalanobis distance is central to the Normal distribution.

- Lets take a moment to examine the general quadratic form:

$$\mathbf{x}^{\mathsf{T}}\mathbf{W}\mathbf{x}$$

  where:

  - first $\mathbf{W}$ is a $d \times d$ full rank symmetric matrix,
  - and then it is a positive semi-definite matrix (PSD), i.e. $\mathbf{x}^T\mathbf{W}\mathbf{x} \geq 0,\ \forall \mathbf{x}$.

A bit more on the Multivariate Normal Distribution, MND II

- Along the way we will examine the properties of the linear transform:

$$\mathbf{A}\mathbf{x},\ \ \mathbf{A} : m \times d, R^d \mapsto R^m$$

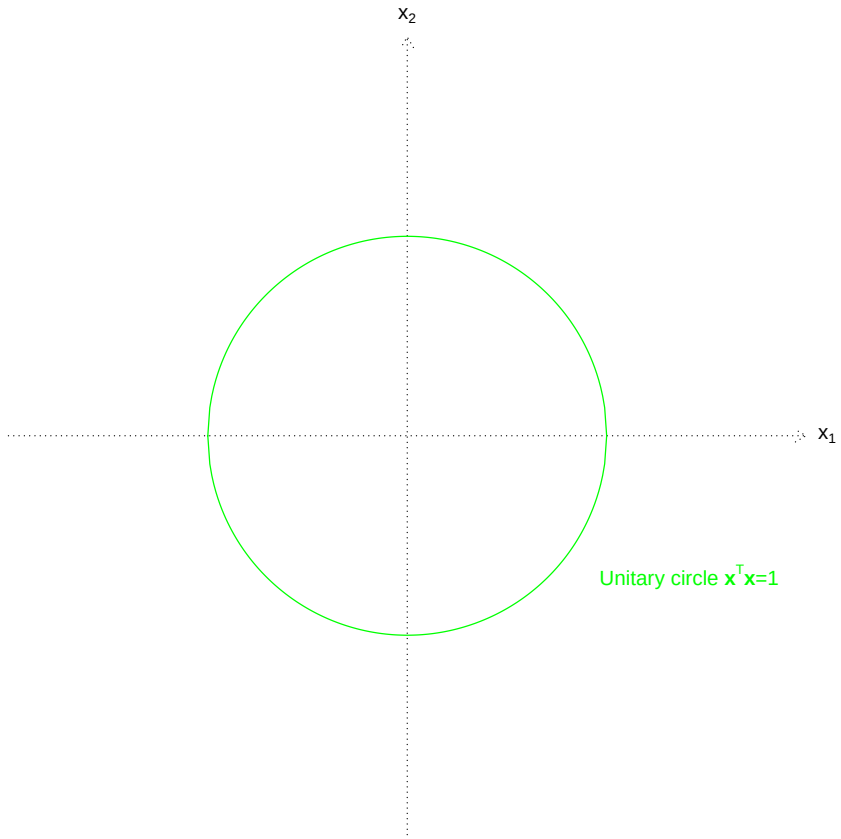
- because the squared norm of the transformed space is:

$$||\mathbf{A}\mathbf{x}||^2 = (\mathbf{A}\mathbf{x})^T\mathbf{A}\mathbf{x} = \mathbf{x}^T\mathbf{A}^T\mathbf{A}\mathbf{x} \overset{\mathbf{W}:=\mathbf{A}}{=} \mathbf{x}^T\mathbf{W}\mathbf{x}$$

- Essentially quadratic forms such as $\mathbf{x}^T\mathbf{W}\mathbf{x}$, where $\mathbf{W}$ can be written as $\mathbf{A}^T\mathbf{A}$ are the squared norms of the $\mathbf{A}\mathbf{x}$ space

Effect of the linear transformation $\mathbf{Ax}$ on the points of the unit circle

- Lets start by exploring the effect of the linear transformation $\mathbf{Ax},\ \mathbf{A} : m \times d, R^d \mapsto R^m$ on the unit circle $\mathbf{x}^T\mathbf{x} = 1$



- demonstration will be done for $d = m$ for simplicity.

Basics on eigen/singular value decomposition

- For square $\mathbf{A}$ of full rank we have the eigen value decomposition

$$\mathbf{A} = \mathbf{U}\mathbf{\Lambda}\mathbf{U}^{-1}$$

columns of $\mathbf{U}$ are the eigenvectors of $\mathbf{A}$, $\lambda_i \geq \lambda_{i+1}$.

- For real symmetric $\mathbf{A}$ we have:

$$\mathbf{A} = \mathbf{U}\mathbf{\Lambda}\mathbf{U}^T$$

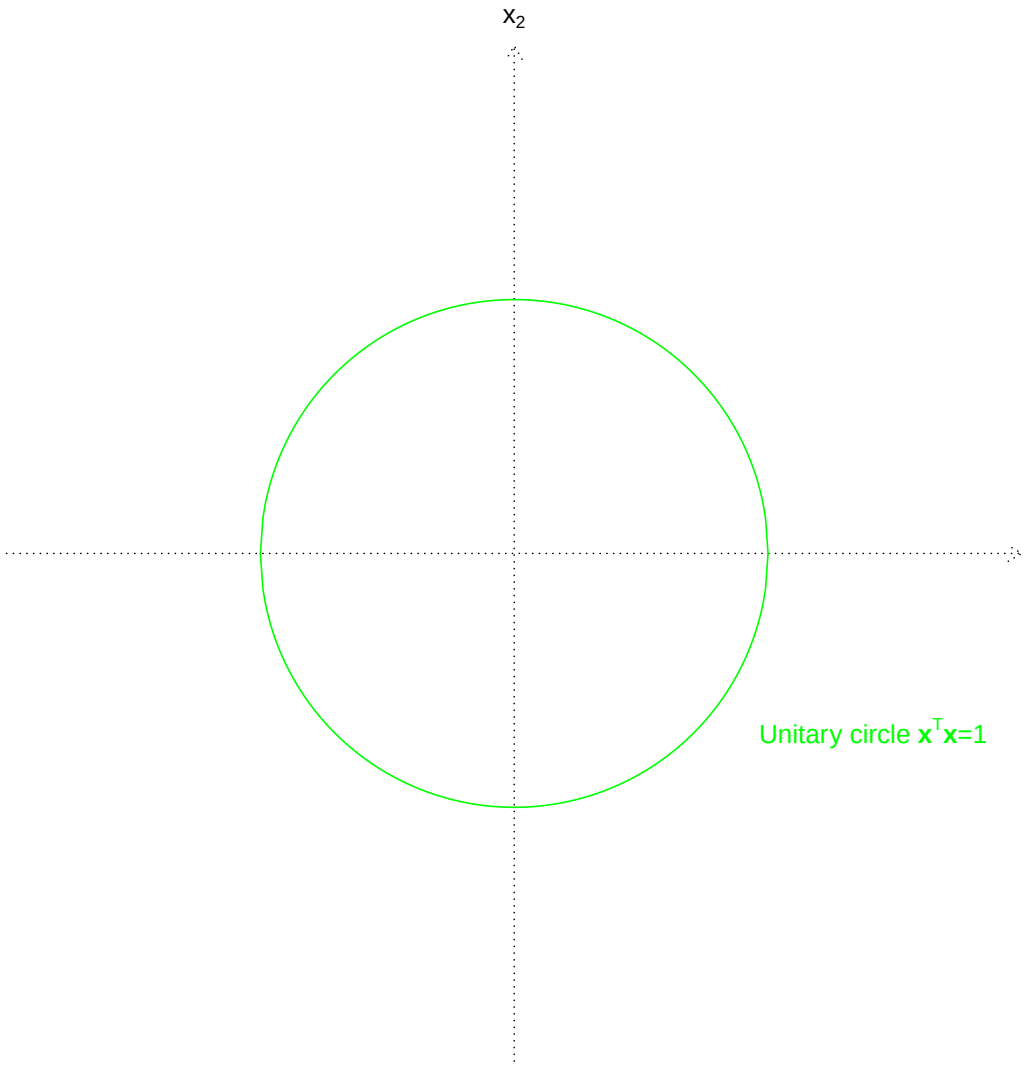
- For rectangular $\mathbf{A}$ we have the singular value decomposition:

$$\mathbf{A} = \mathbf{U}\mathbf{\Sigma}\mathbf{V}^\mathsf{T}$$

  - columns of $\mathbf{U} : m \times m$ are eigenvectors of $\mathbf{A}\mathbf{A}^\mathsf{T}$
  - columns of $\mathbf{V} : d \times d$ are eigenvectors of $\mathbf{A}^\mathsf{T}\mathbf{A}$
  - the diagonal element σ_i of the $\mathbf{\Sigma} : m \times d$ matrix is the square root of the λ_i eigenvalue of $\mathbf{A}^T\mathbf{A}$, $\mathbf{A}\mathbf{A}^\mathsf{T}$

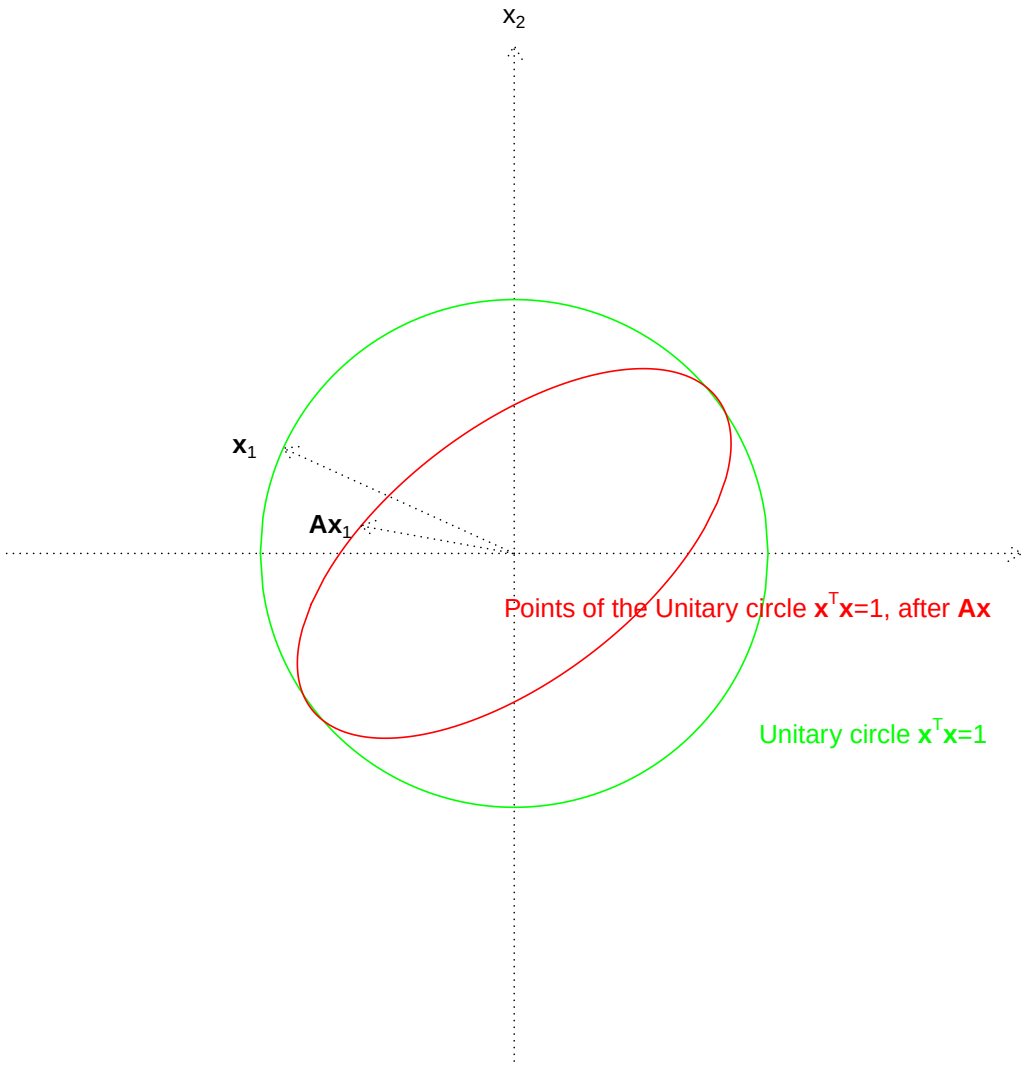
Effect of the linear transformation $\mathbf{Ax}$ on the points of the unit circle

- We will work with the square $\mathbf{A} = \begin{bmatrix} 0.8 & 0.3 \\ 0.2 & 0.7 \end{bmatrix}$
  - that has eigenvectors and eigenvalues:
    - $\mathbf{u}_1 = (0.83, 0.55)^\mathsf{T},\ \lambda_1 = 1$
    - $\mathbf{u}_2 = (-0.71, 0.71),\ \lambda_1 = 0.5$



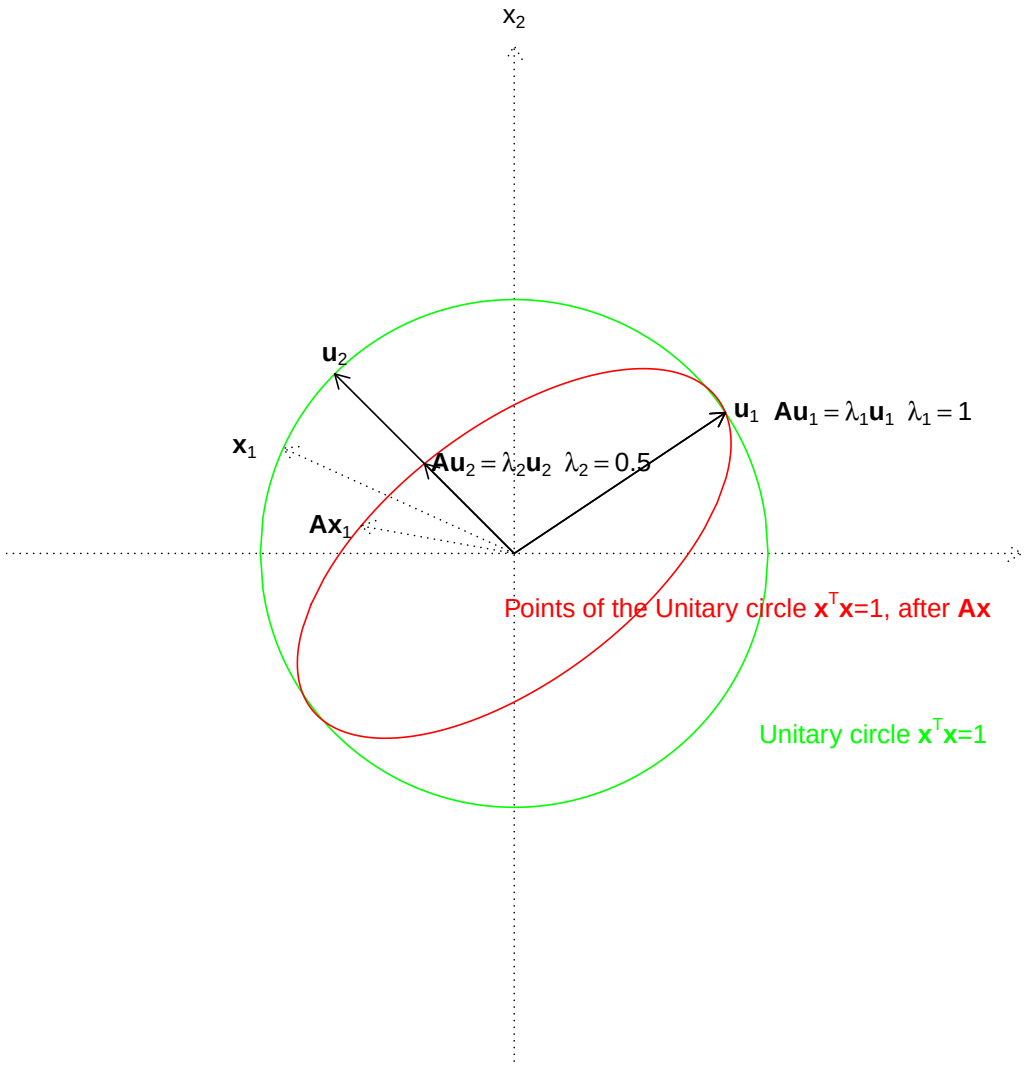
Effect of the linear transformation $\mathbf{Ax}$ on the points of the unit circle

- $\mathbf{Ax}$ transforms the unit circle to an ellipse, squezing or enlarging distances from the origin.
- For any general vector $\mathbf{x}$ the linear transformation $\mathbf{Ax}$ changes its direction
- except for...



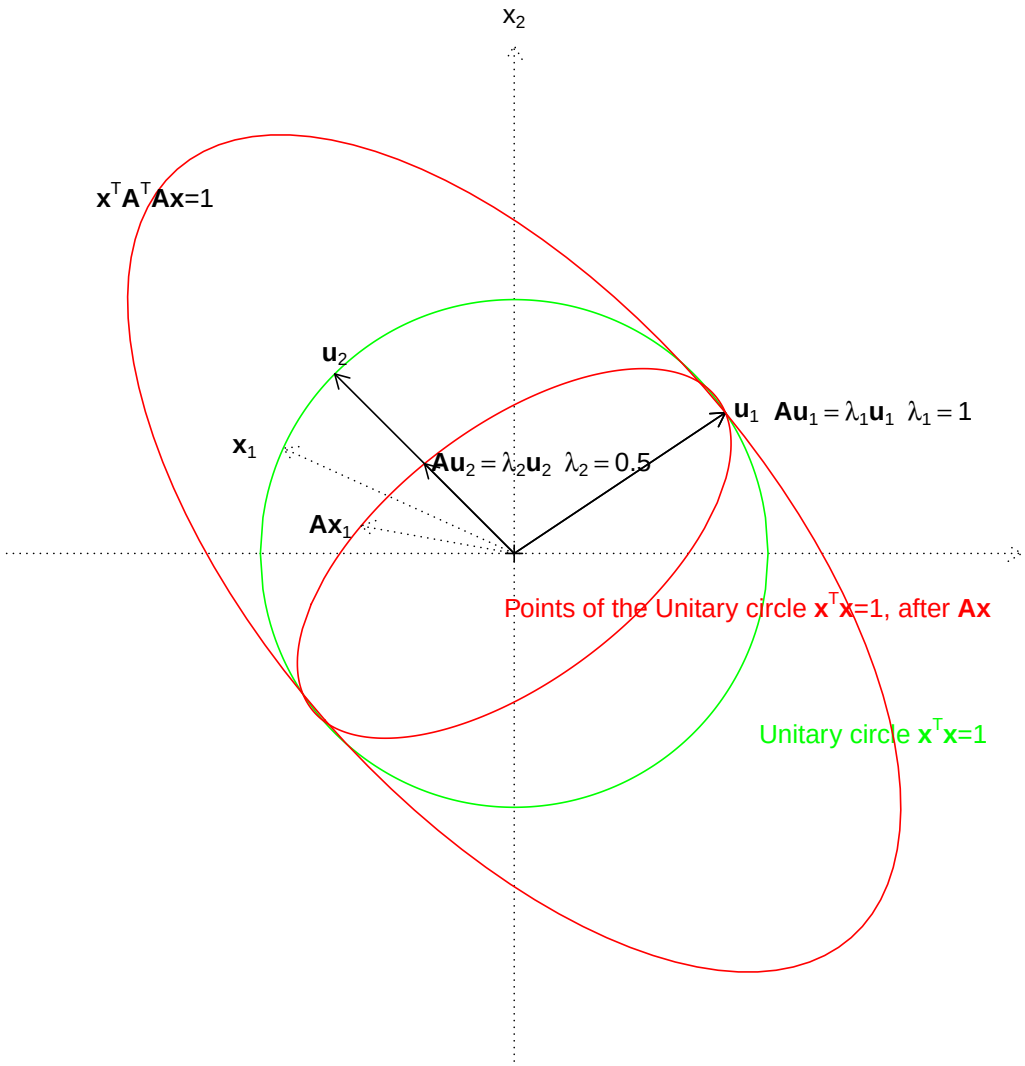
Effect of the linear transformation $\mathbf{Ax}$ on the points of the unit circle

- ... the eigenvectors $\mathbf{u}_1, \mathbf{u}_2$, of $\mathbf{A}$ which are the only vectors whose direction does not change by $\mathbf{A}$.
- They are only scaled by the λ_1 and λ_2 eigenvalues respectively.
- $||\mathbf{Au}_i||^2 = (\mathbf{Au}_i)^\mathsf{T}(\mathbf{Au}_i) = (\lambda_i\mathbf{u}_i)^\mathsf{T}(\lambda_i\mathbf{u}_i) = \lambda_i^2||\mathbf{u}_i||^2$



Effect of the linear transformation $\mathbf{Ax}$ on the points of the unit circle

- Distance from the origin, $||\mathbf{Ax}||$, evolves much slower on the direction of $\mathbf{u}_2$ compared to $\mathbf{u}_1$.



On the $\mathbf{x}^T\mathbf{W}\mathbf{x}$ quadratic form, I

- If $\mathbf{W}$ is *symmetric* matrix $\mathbf{W}$:
  - All its eigenvalues are *real*.
  - Eigenvectors of different eigenvalues are *orthogonal*[1] to each other.
- In addition $\mathbf{W}$ can be decomposed as:

$$\mathbf{W} = \mathbf{V}\mathbf{\Lambda}\mathbf{V}^T \tag{3}$$

  (spectral theorem), where
  - $\mathbf{V}$ is the orthogonal matrix with columns the orthonormal eigenvectors, $\mathbf{v}_i, i = 1 \ldots d$, of $\mathbf{W}$,
  - and $\mathbf{\Lambda}$ the diagonal eigen values matrix.

On the $\mathbf{x}^T\mathbf{W}\mathbf{x}$ quadratic form, II

- Then:

$$\begin{aligned} \mathbf{x}^T\mathbf{W}\mathbf{x} &= \mathbf{x}^T\mathbf{V}\mathbf{\Lambda}\mathbf{V}^T\mathbf{x} \\ &= [\mathbf{V}^T\mathbf{x}]^T\mathbf{\Lambda}\mathbf{V}^T\mathbf{x} \\ &= \mathbf{y}^T\mathbf{\Lambda}\mathbf{y} \end{aligned} \tag{4}$$

- The linear transformation $\mathbf{y} := \mathbf{V}^T\mathbf{x}$ does ???

On the $\mathbf{x}^T\mathbf{W}\mathbf{x}$ quadratic form, II

- Then:

$$\begin{aligned} \mathbf{x}^T\mathbf{W}\mathbf{x} &= \mathbf{x}^T\mathbf{V}\mathbf{\Lambda}\mathbf{V}^T\mathbf{x} \\ &= [\mathbf{V}^T\mathbf{x}]^T\mathbf{\Lambda}\mathbf{V}^T\mathbf{x} \\ &= \mathbf{y}^T\mathbf{\Lambda}\mathbf{y} \end{aligned} \tag{4}$$

- The linear transformation $\mathbf{y} := \mathbf{V}^T\mathbf{x}$ does ??? a projection of $\mathbf{x}$ to the eigenvector space of $\mathbf{W}$ (a change of basis).
- Moreover:

$$\mathbf{y}^T\mathbf{\Lambda}\mathbf{y} = \sum y_i^2\lambda_i$$

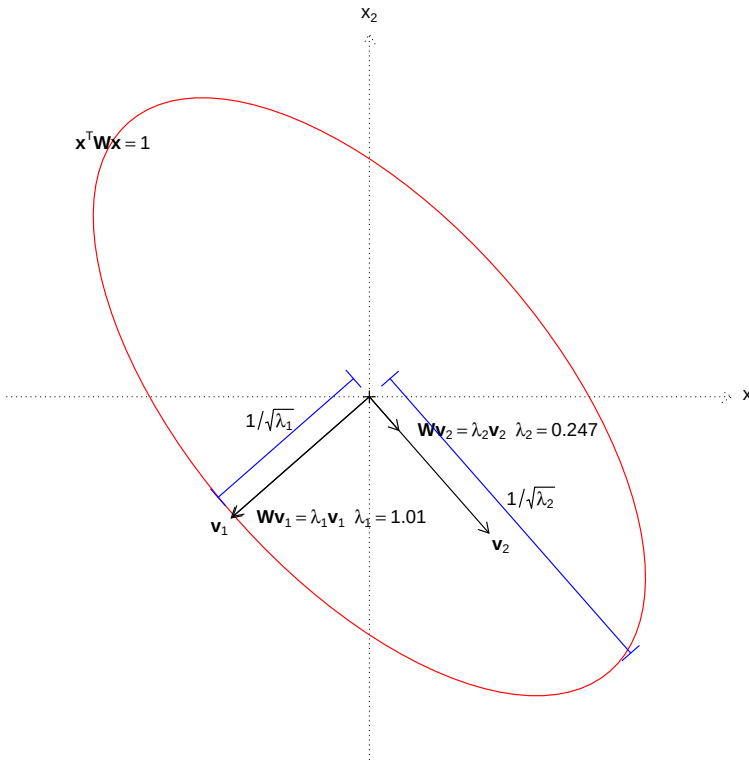
- If in addition $\mathbf{W}$ is PSD, then it has $\lambda_i \geq 0$ and we can write

$$\mathbf{y}^T\mathbf{\Lambda}\mathbf{y} = \sum y_i^2\lambda_i = \sum(\frac{y_i}{1/\sqrt{\lambda_i}})^2 \tag{5}$$

which is...

On the $\mathbf{x}^T\mathbf{W}\mathbf{x}$ quadratic form, III

- a hyperelipsoid centered at the origin of the axes:
  - the direction of its main axes is given by the eigenvectors $\mathbf{v}_i$ of $\mathbf{W}$
  - and the scaling of the axes by $\frac{1}{\sqrt{\lambda_i}}$ (λ_i the eigen values of $\mathbf{W}$)
- if $\mathbf{W}$ is only symmetric and not PSD then $\mathbf{y}^T\mathbf{\Lambda}\mathbf{y} = \sum y_i^2\lambda_i$ is a hyperbola.
- (for the figure we set: $\mathbf{W} = \mathbf{A}^T\mathbf{A}$, where $\mathbf{A} = \begin{bmatrix} 0.8 & 0.3 \\ 0.2 & 0.7 \end{bmatrix}$)



On the $\mathbf{x}^T\mathbf{W}\mathbf{x}$ quadratic form, III

- a hyperelipsoid centered at the origin of the axes:
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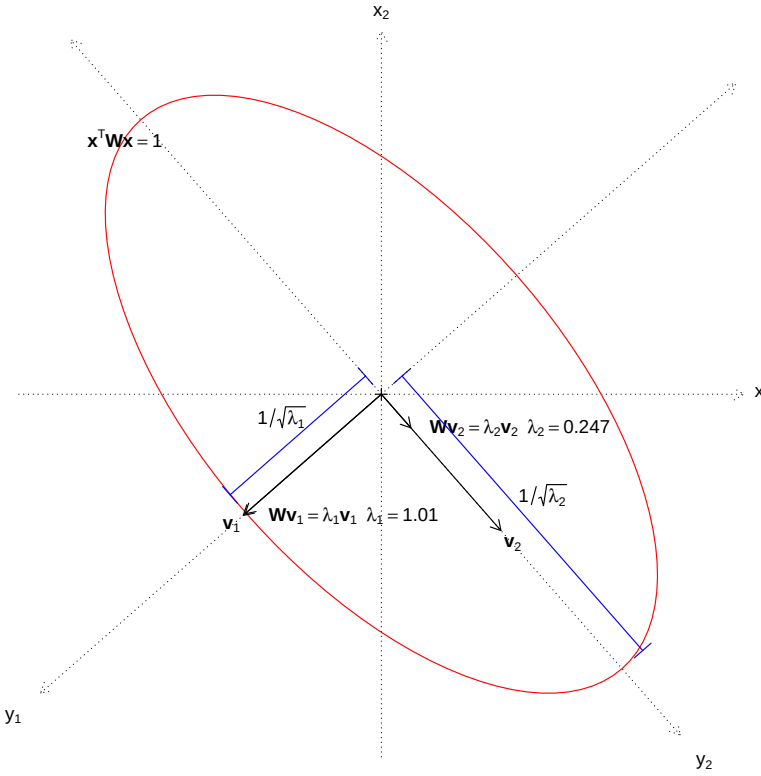
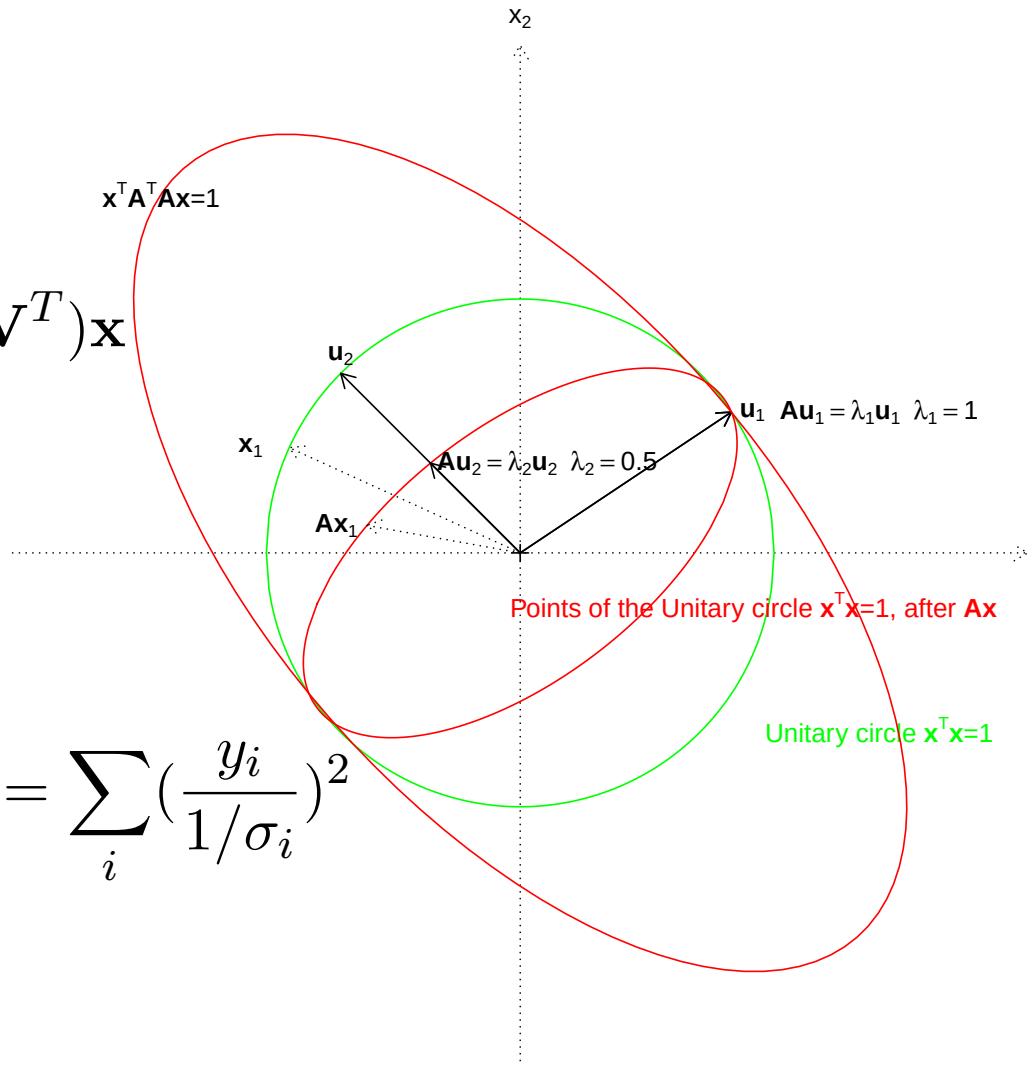
- (for the figure we set: $\mathbf{W} = \mathbf{A}^T\mathbf{A}$, where $\mathbf{A} = \begin{bmatrix} 0.8 & 0.3 \\ 0.2 & 0.7 \end{bmatrix}$)
- $\mathbf{y} := \mathbf{V}^T\mathbf{x}$ does the projection on the new space given by the main axes of the ellipsoid.



On the relation of $\mathbf{Ax}$ and the $\mathbf{x}^T\mathbf{A}^T\mathbf{Ax}$ quadratic form

- Let $\mathbf{A} = \mathbf{U\Sigma V}^T$ the singular value decomposition of $\mathbf{A}$

$$\begin{aligned}\mathbf{x}^T\mathbf{A}^T\mathbf{Ax} &= \mathbf{x}^T(\mathbf{U\Sigma V}^T)^T(\mathbf{U\Sigma V}^T)\mathbf{x} \\ &= \mathbf{x}^T\mathbf{V}^T\mathbf{\Sigma}^T\mathbf{\Sigma V x}\end{aligned}$$



Let $\mathbf{y} := \mathbf{V}^T\mathbf{x}$ then:

$$\mathbf{x}^T\mathbf{A}^T\mathbf{Ax} = \mathbf{y}^T\mathbf{\Sigma}^T\mathbf{\Sigma y} = \sum_i {y_i}^2\sigma_i^2 = \sum_i (\frac{y_i}{1/\sigma_i})^2$$

- which the same as eq 5 but written with the singular values of $\mathbf{A}$.

On the $(\mathbf{x}-\boldsymbol{\mu})^T\boldsymbol{\Sigma}^{-1}(\mathbf{x}-\mu)$ quadratic form, I

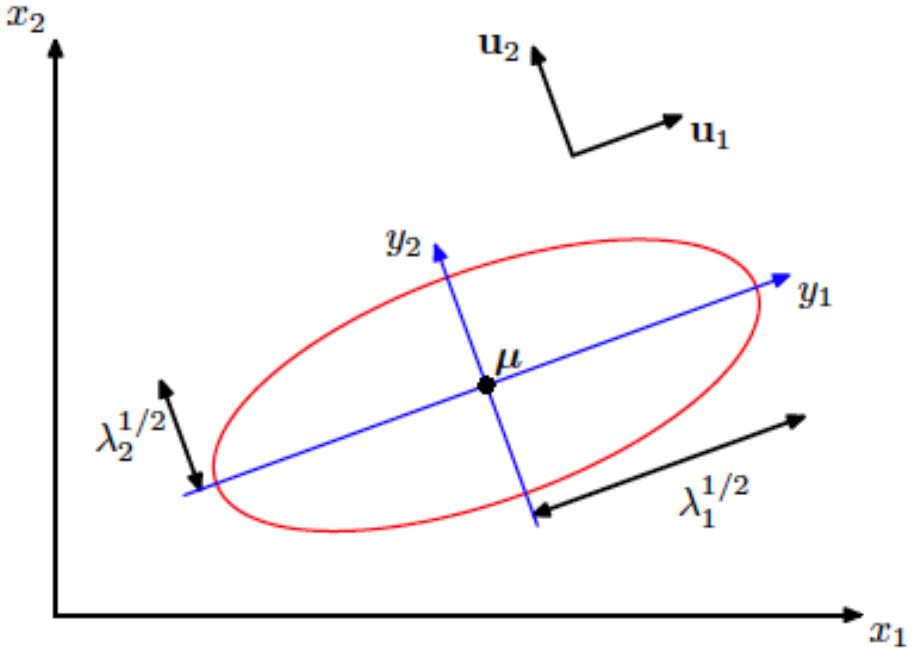
- Let the eigen decomposition of $\boldsymbol{\Sigma} = \mathbf{U}\boldsymbol{\Lambda}\mathbf{U}^T$
- the decomposition of its inverse is $\boldsymbol{\Sigma}^{-1} = \mathbf{U}\boldsymbol{\Lambda}^{-1}\mathbf{U}^T$
- then:

$$\begin{aligned}(\mathbf{x}-\boldsymbol{\mu})^T\boldsymbol{\Sigma}^{-1}(\mathbf{x}-\mu) &= (\mathbf{x}-\boldsymbol{\mu})^T\mathbf{U}\boldsymbol{\Lambda}^{-1}\mathbf{U}^T(\mathbf{x}-\boldsymbol{\mu}) \qquad (6)\\ &= [\mathbf{U}^T(\mathbf{x}-\boldsymbol{\mu})]^T\boldsymbol{\Lambda}^{-1}[\mathbf{U}^T(\mathbf{x}-\boldsymbol{\mu})]]\\ &= \mathbf{y}^T\boldsymbol{\Lambda}^{-1}\mathbf{y}\\ &= \sum_i(\frac{y_i}{\sqrt{\lambda_i}})^2\end{aligned}$$

- The transformation $\mathbf{U}^T(\mathbf{x}-\boldsymbol{\mu})$ translates by $\boldsymbol{\mu}$ and projects on the eigenvectors.
- $\sum_i(\frac{y_i}{\sqrt{\lambda_i}})^2$ is...

On the $(\mathbf{x}-\boldsymbol{\mu})^T\boldsymbol{\Sigma}^{-1}(\mathbf{x}-\mu)$ quadratic form, II

- ...



- A hyperelipsoid with:
  - the directions of its main axes given by the eigenvectors, $\mathbf{u}_i$ of the covariance matrix $\boldsymbol{\Sigma}$
  - and the scaling of the axes given by the eigenvalues $\sqrt{\lambda_i}$
- Thus $\mathbf{U}^T(\mathbf{x}-\boldsymbol{\mu})$ translates the origin to $\boldsymbol{\mu}$ and aligns the axes with the main axes of the hyperelipsoid.

Principal component analysis

- Trace of $\mathbf{\Sigma} = \sum_i \sigma_i^2 = \sum_i \lambda_i =$ Total Variance
- In Principal Component Analysis, PCA (a feature transformation method), we project instances on $\mathbf{U}^T\mathbf{x}$.
- Each projection dimension

$$y_i = \mathbf{u}_i^T\mathbf{x}$$

  gives the projection on the ith eigen vector,
- the variance of y_i is λ_i
- the y_is are decorelated
- y_is are ordered by λ_i
- PCA can be used as dimensionality reduction (keep the $\mathbf{u}_i$s with largest λ_i)

PCA and $(\mathbf{x}-\boldsymbol{\mu})^\mathsf{T}\boldsymbol{\Sigma}^{-1}(\mathbf{x}-\mu)$

- We saw that the Mahalanobis distance in the Normal distribution transforms data as:

$$(\mathbf{x}-\boldsymbol{\mu})^\mathsf{T}\boldsymbol{\Sigma}^{-1}(\mathbf{x}-\mu) = [\mathbf{U}^\mathsf{T}(\mathbf{x}-\boldsymbol{\mu})]^\mathsf{T}\boldsymbol{\Lambda}^{-1}[\mathbf{U}^\mathsf{T}(\mathbf{x}-\boldsymbol{\mu})]]$$

$$\mathbf{y} := \mathbf{U}^\mathsf{T}(\mathbf{x}-\boldsymbol{\mu})]^\mathsf{T}\sqrt{\Lambda}^{-1}$$

  i.e. new dimensions y_i have the mean sustracted, are decorelated, and have a variance of 1.

- in PCA the transformation is

$$\mathbf{y} := \mathbf{U}^\mathsf{T}\mathbf{x}$$

  i.e. new dimensions y_i are decorelated and have a variance of λ_i.

Back to LD using distributional assumptions

- Remember we stopped at:

$$\begin{aligned}\arg\max_{c_i}(P(c_i|\mathbf{x})) &= \arg\max_{c_i}[\ln(\frac{1}{(2\pi)^{d/2}|\boldsymbol{\Sigma}_i|^{1/2}}) \\ &\quad -\frac{1}{2}(\mathbf{x}-\boldsymbol{\mu}_i)^{\mathsf{T}}\boldsymbol{\Sigma}_i^{-1}(\mathbf{x}-\boldsymbol{\mu_i}) \\ &\quad +\ln(P(c_i))]\end{aligned}$$

- Depending on the form of the covariance matrices we get different classifiers.

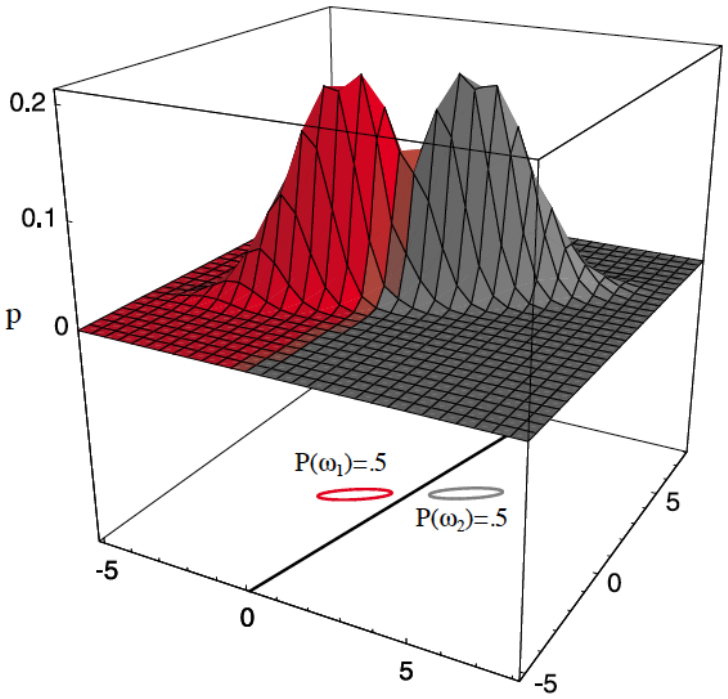
LD using distributional assumptions, $\Sigma_i = \Sigma$

- All classes have the same covariance Σ.
- Then the term, $\ln(\frac{1}{(2\pi)^{d/2}|\Sigma|^{1/2}})$, is the same for all classes. So:

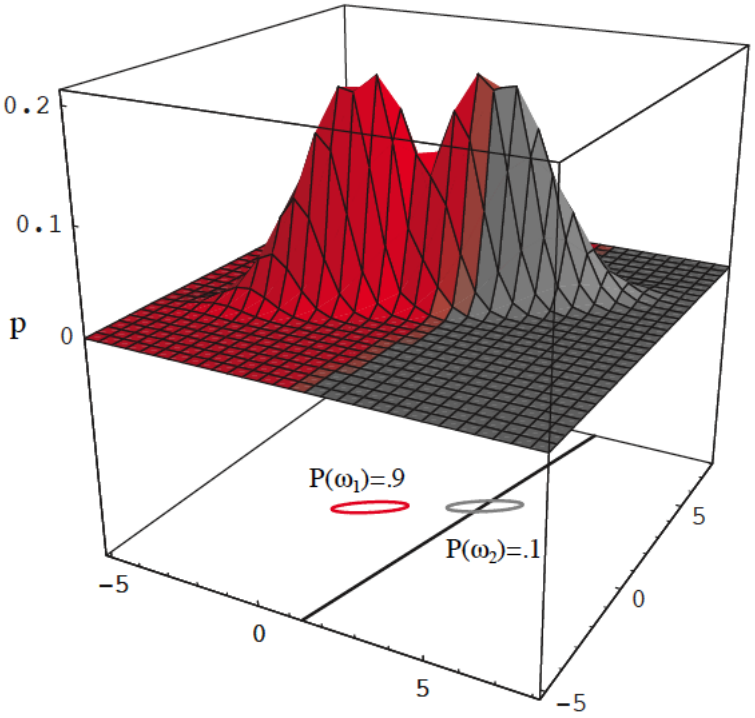
$$\begin{aligned} \arg\max_{c_i}(P(c_i|\mathbf{x})) &= \arg\max_{c_i}[-\frac{1}{2}(\mathbf{x}-\mu_\mathbf{i})^\mathsf{T}\Sigma^{-1}(\mathbf{x}-\mu_\mathbf{i}) \\ &\quad + \ln(P(c_i))] \end{aligned} \tag{7}$$

- So the classification rule is a simple function of the *Mahalanobis* distance from the class centers and the class priors.
- If in equation 7 you develop the Mahalanobis distance you will see that this is a linear function with respect to $\mathbf{x}$, (the quadratic term $\mathbf{x}^t\Sigma^{-1}\mathbf{x}$ is independent of the class and can be omitted).

LD using distributional assumptions, $\Sigma_i = \Sigma$. Examples I

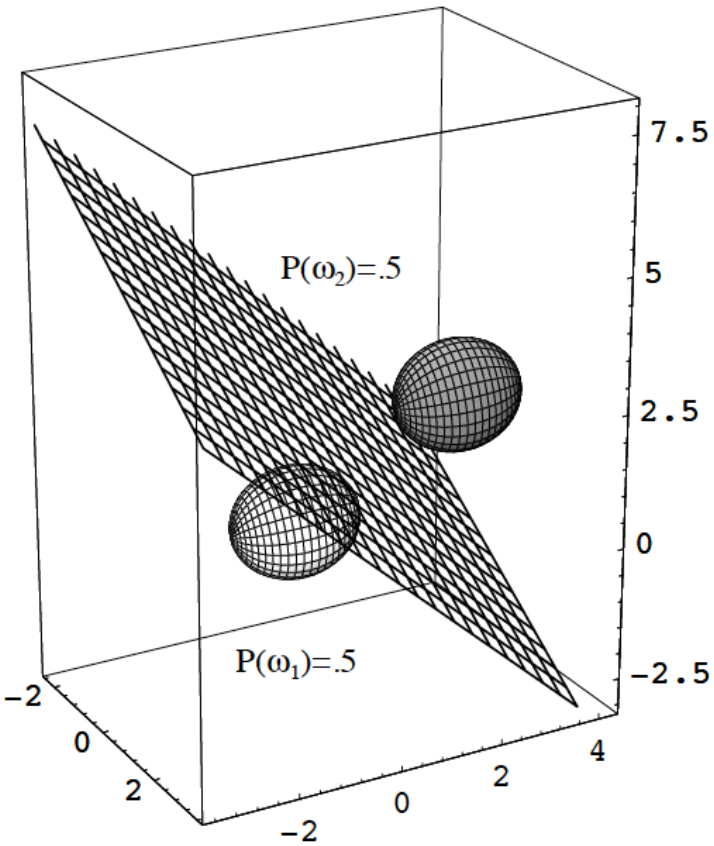


Equal class priors

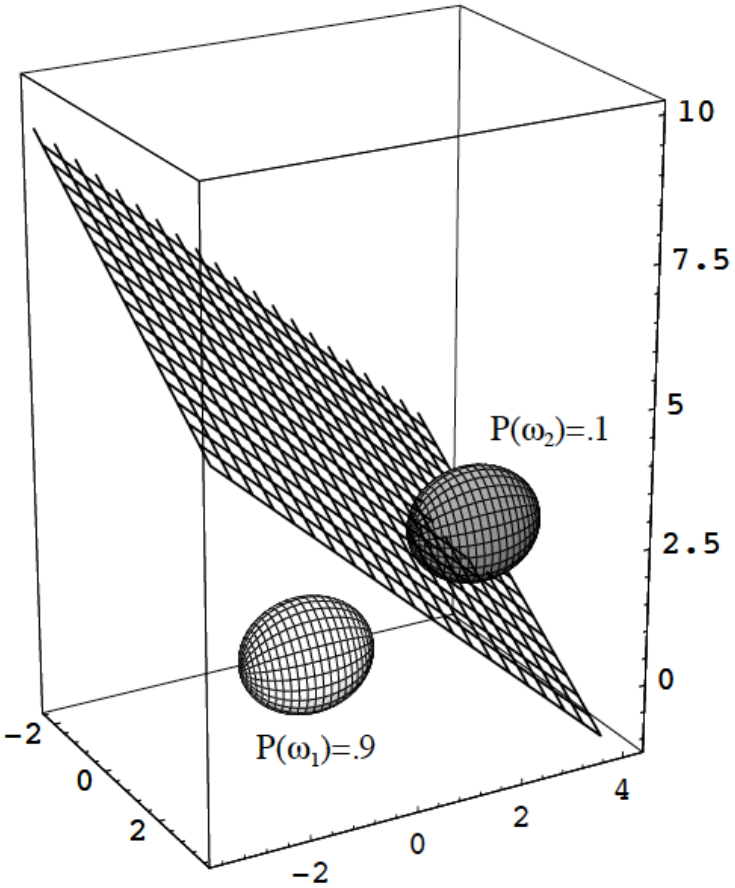


Unequal class priors

LD using distributional assumptions, $\Sigma_i = \Sigma$. Examples II



Equal class priors



Unequal class priors

LD using distributional assumptions, $\mathbf{\Sigma}_i = \sigma^2 \mathbf{I}$

- In this case:

$$-\frac{1}{2}(\mathbf{x} - \boldsymbol{\mu}_i)^t \mathbf{\Sigma}_i^{-1} (\mathbf{x} - \boldsymbol{\mu}_i) =$$

LD using distributional assumptions, $\mathbf{\Sigma}_i = \sigma^2\mathbf{I}$

- In this case:

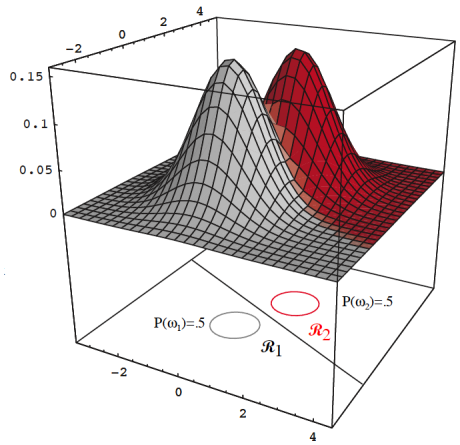
$$-\frac{1}{2}(\mathbf{x}-\boldsymbol{\mu}_i)^t\mathbf{\Sigma}_i^{-1}(\mathbf{x}-\boldsymbol{\mu}_i) = -\frac{||\mathbf{x}-\boldsymbol{\mu}_i||^2}{2\sigma^2}$$

so:

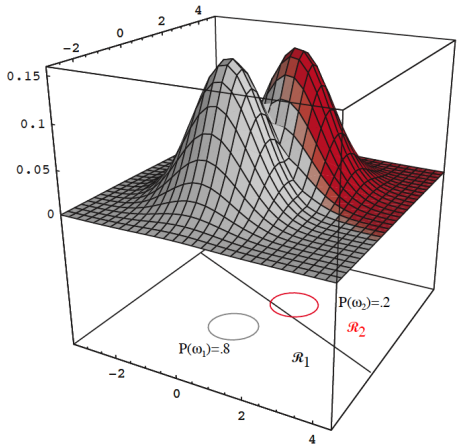
$$\begin{aligned}\arg\max_{c_i}(P(c_i|\mathbf{x})) &= \arg\max_{c_i}[-\frac{||\mathbf{x}-\boldsymbol{\mu}_i||^2}{2\sigma^2} \\ &\quad + \ln(P(c_i))]\end{aligned}$$

- So the classification rule is a simple function of the *Euclidean* distance from the class centers and the class priors.
- Again developing the function we can see that it is linear with respect to $\mathbf{x}$.

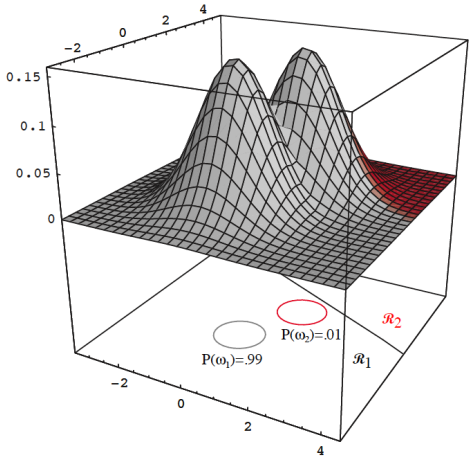
LD using distributional assumptions, $\boldsymbol{\Sigma}_i = \sigma^2 \boldsymbol{I}$



Equal class priors



Unequal class priors



Very unequal class priors

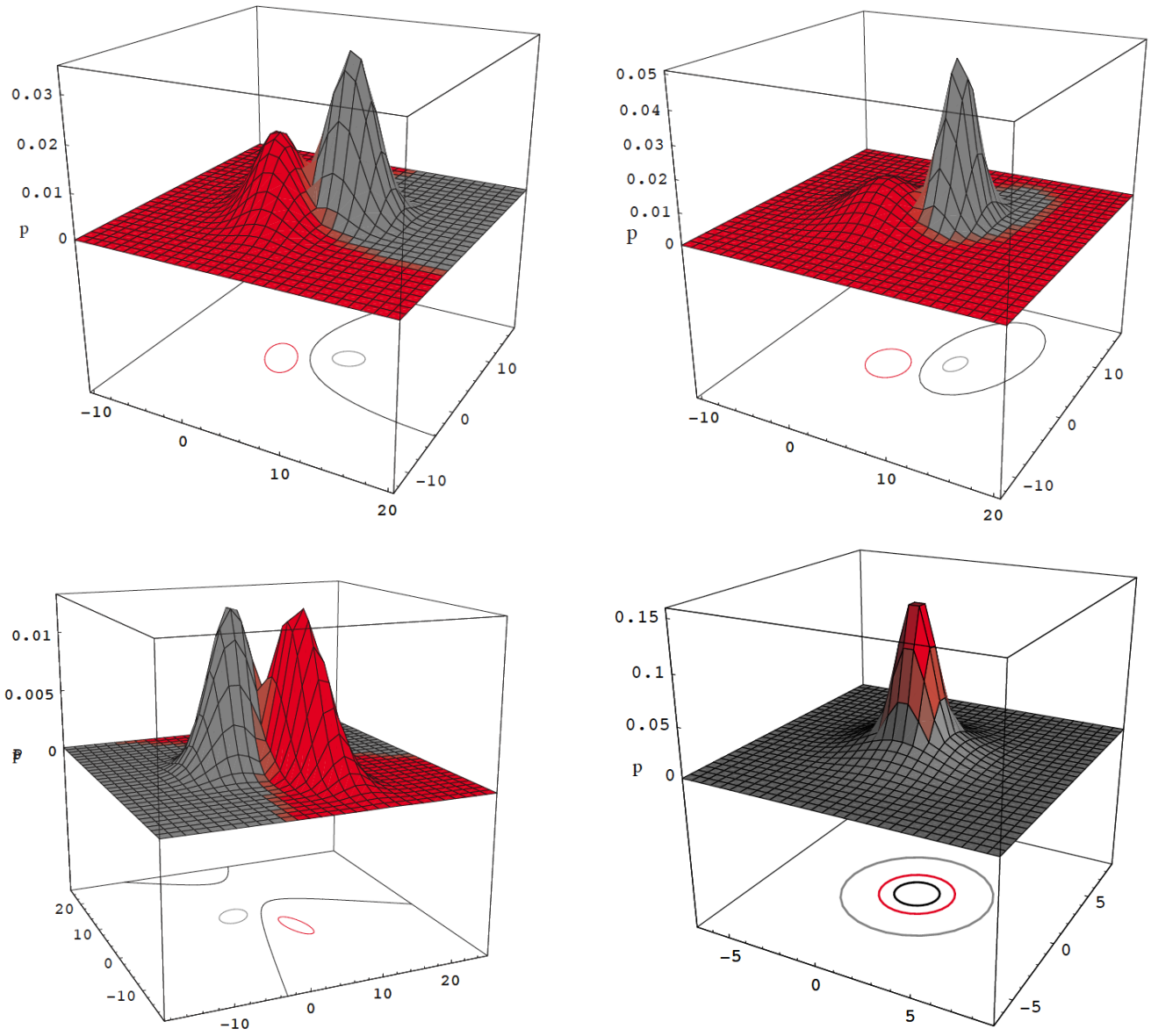
LD using distributional assumptions, arbitrary Σ_i

- In this case no simplifications can be made and the classification rule remains:

$$\begin{aligned}\arg\max_{c_i}(P(c_i|\mathbf{x})) &= \arg\max_{c_i}[\ln(\frac{1}{(2\pi)^{d/2}|\boldsymbol{\Sigma_i}|^{1/2}}) \\ &\quad -\frac{1}{2}(\mathbf{x}-\mu_\mathbf{i})^t\boldsymbol{\Sigma}_i^{-1}(\mathbf{x}-\mu_\mathbf{i}) \\ &\quad +\ln(P(c_i))]\end{aligned} \tag{8}$$

  which is a *quadratic* function of $\mathbf{x}$.
- The decision surfaces are hyperquadrics, and can assume any of the general forms: hyperplanes, pairs of hyperplanes, hyperspheres, hyperellipsoids, hyperparaboloids, quadric and hyperhyperboloids of various types

LD using distributional assumptions, arbitrary Σ_i

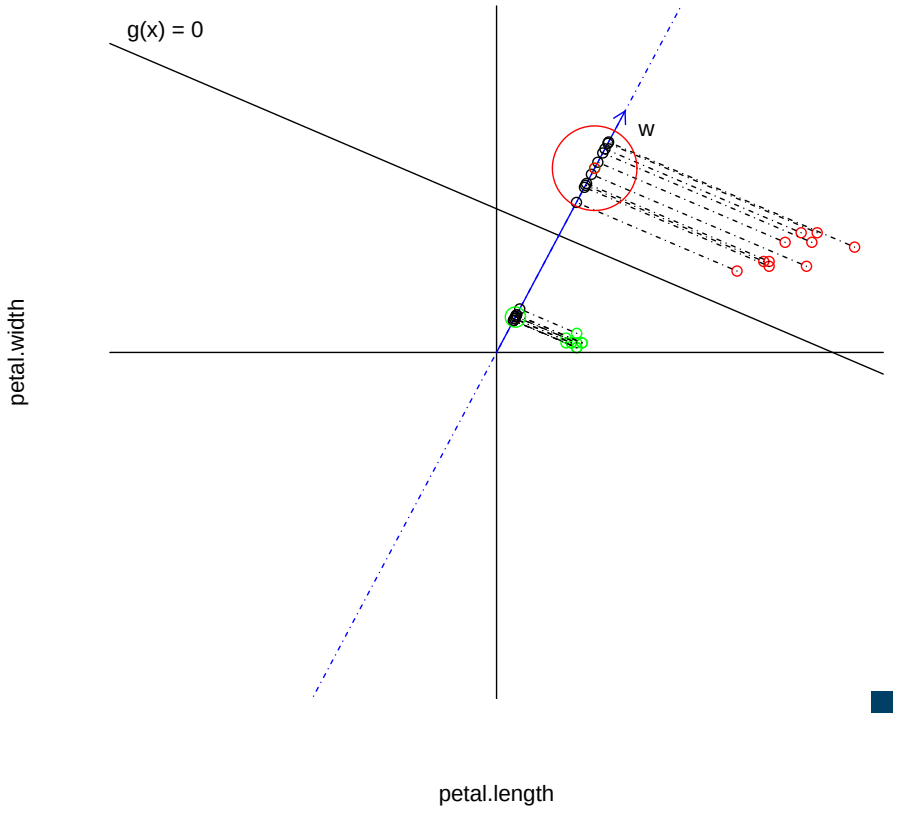


Fisher's Linear Discriminant

Fisher's Linear Discriminant

- Project $\mathbf{x}$ from the initial d-dimensional space on to a single line, i.e., a 1-dimensional space, in case of two classes, or to a $|c| - 1$ dimensional space in case of $|c|$ classes.
- The direction $\mathbf{w}$ of the line should be chosen in such a way so that the separation will be maximum.

Fisher's linear discriminants: Intuition



- Fisher's linear discriminants will look for that projection direction $\mathbf{w}$ that:
  - minimizes the sum of the radius of the two circles, i.e.

$$\min(R_{red} + R_{green})$$

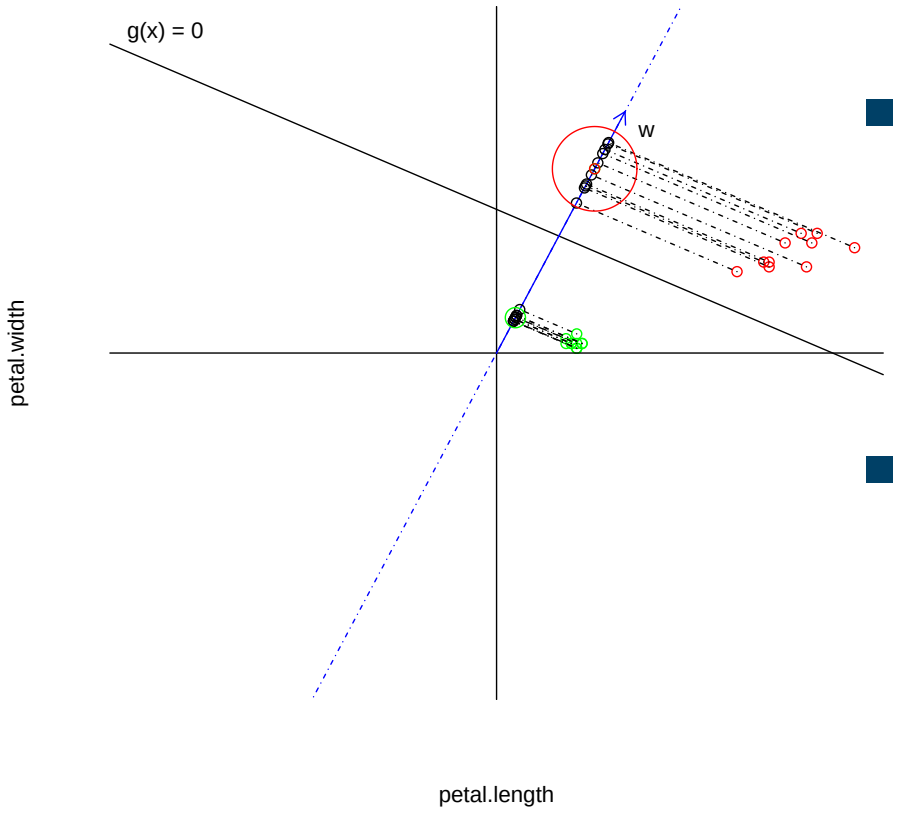
  - places the centers of the two circles, C_{red}, C_{green}, as far as possible from each other, i.e.

$$\max(|C_{red} - C_{green}|)$$

- it achieves the above by looking for the direction $\mathbf{w}$ that maximizes:

$$\frac{|C_{red} - C_{green}|}{R_{red} + R_{green}}$$

Fisher's linear discriminants: Intuition



- How can we define the centers C_{red}, C_{green}?
  - We will call the distance $|C_{red} - C_{green}|$ *between class distance*
- How can we define the radius R_{red}, R_{green}?
  - We will call the $|R_{red} + R_{green}|$ *within class distance*

Fisher's linear discriminants: Intuition I

- We will work in a two-class setting, thus we project to one dimension, thus projections are just scalars.
- Let $\bar{x}$ denote the Fisher projection of $\mathbf{x}$ instance
- and $\bar{\mu}_i$ denote the Fisher projection of the $\boldsymbol{\mu}_i$ mean of the i class.
- We need measures of:
  - of the distances of the classes in the projected space, *between class distance*
  - of the spread of the instances of each class in the projected space, *within class distance*

Fisher's linear discriminants: Between class distance

- We can measure the *between class distance* in the projected space by...
- ...the squared distance of the projections of the means:

$$(\bar{\mu_1} - \bar{\mu_2})^2 \quad =$$

Fisher's linear discriminants: Between class distance

- We can measure the *between class distance* in the projected space by...
- ...the squared distance of the projections of the means:

$$
\begin{aligned}
(\bar{\mu_1} - \bar{\mu_2})^2 &= (\frac{1}{n_1}\sum_{\mathbf{x_i} \in c_1} \mathbf{w}^T \mathbf{x_i} - \frac{1}{n_2}\sum_{\mathbf{x_i} \in c_2} \mathbf{w}^T \mathbf{x_i})^2 \\
&= (\mathbf{w}^T \boldsymbol{\mu}_1 - \mathbf{w}^T \boldsymbol{\mu}_2)^2 \\
&= \mathbf{w}^T (\boldsymbol{\mu}_1 - \boldsymbol{\mu}_2)[\mathbf{w}^T (\boldsymbol{\mu}_1 - \boldsymbol{\mu}_2)]^T \\
&= \mathbf{w}^T (\boldsymbol{\mu}_1 - \boldsymbol{\mu}_2)(\boldsymbol{\mu}_1 - \boldsymbol{\mu}_2)^T \mathbf{w} \\
&= \mathbf{w}^T \mathbf{S_B} \mathbf{w}
\end{aligned}
$$

  which is a measure of how far are the two classes.
- $\mathbf{S_B}$ is called the *between class scatter* matrix, $d \times d$, and is of rank one.

Fisher's linear discriminants: Within class distance

- We need measures of the spread of the instances of each class in the projected space, *within class distance*,
- by the variance of the projections around their respective means

$$
\begin{aligned}
\bar{s}_i^2 &= \sum_{\mathbf{x} \in C_i} (\mathbf{w}^T \mathbf{x} - \mathbf{w}^T \boldsymbol{\mu}_i)^2 \\
&= \sum_{\mathbf{x} \in C_i} [\mathbf{w}^T (\mathbf{x} - \boldsymbol{\mu}_i)(\mathbf{x} - \boldsymbol{\mu}_i)^T \mathbf{w}] \\
&= \mathbf{w}^T [\sum_{\mathbf{x} \in C_i} (\mathbf{x} - \boldsymbol{\mu}_i)(\mathbf{x} - \boldsymbol{\mu}_i)^T] \mathbf{w} \\
&= \mathbf{w}^T \mathbf{S}_i \mathbf{w}
\end{aligned}
$$

- $\mathbf{S}_i$ is the *within class scatter* matrix of the i class, and what else????

Fisher's linear discriminants: Within class distance

- We need measures of the spread of the instances of each class in the projected space, *within class distance*,
- by the variance of the projections around their respective means

$$
\begin{aligned}
\bar{s}_i^2 &= \sum_{\mathbf{x} \in C_i} (\mathbf{w}^T \mathbf{x} - \mathbf{w}^T \boldsymbol{\mu}_i)^2 \\
&= \sum_{\mathbf{x} \in C_i} [\mathbf{w}^T (\mathbf{x} - \boldsymbol{\mu}_i)(\mathbf{x} - \boldsymbol{\mu}_i)^T \mathbf{w}] \\
&= \mathbf{w}^T [\sum_{\mathbf{x} \in C_i} (\mathbf{x} - \boldsymbol{\mu}_i)(\mathbf{x} - \boldsymbol{\mu}_i)^T] \mathbf{w} \\
&= \mathbf{w}^T \mathbf{S}_i \mathbf{w}
\end{aligned}
$$

- $\mathbf{S}_i$ is the *within class scatter* matrix of the i class, and what else????
- .. it is just $(n_i - 1)\mathbf{C}_i$, where $\mathbf{C}_i$ the covariance matrix of the i class.

Fisher's Linear Discriminant, cost function

- So to achieve the intuitive rule we describe Fisher's linear discriminant $\mathbf{w}$ according to:

$$\begin{aligned}
\arg\max_{\mathbf{w}} J(\mathbf{w}) &= \arg\max_{\mathbf{w}} \frac{(\bar{\mu_1} - \bar{\mu_2})^2}{\bar{s}_1^2 + \bar{s}_2^2} \\
&= \arg\max_{\mathbf{w}} \frac{\mathbf{w}^T \mathbf{S_B} \mathbf{w}}{\mathbf{w}^T (\mathbf{S}_1 + \mathbf{S}_2) \mathbf{w}} \\
&= \arg\max_{\mathbf{w}} \frac{\mathbf{w}^T \mathbf{S_B} \mathbf{w}}{\mathbf{w}^T \mathbf{S_W} \mathbf{w}} \\
&= \arg\max_{\mathbf{w}} \frac{between\ class\ scatter}{within\ class\ scatter}
\end{aligned} \tag{9}$$

- $\mathbf{S}_W = \mathbf{S}_1 + \mathbf{S}_2$ is the within class scatter matrix.
- note that FLD does not compute w_0 we will see how this can be done.

Optimizing the Fisher's Linear Discriminant cost function I

- One way to maximize $J(\mathbf{w})$ of eq 9 is:
  - differentiate $J(\mathbf{w})$ with respect to $\mathbf{w}$ and set it to zero, by which we get:

$$(\mathbf{w}^T \mathbf{S}_B \mathbf{w})\mathbf{S_W}\mathbf{w} \quad = \quad \mathbf{S_B}\mathbf{w}(\mathbf{w}^T \mathbf{S_W}\mathbf{w})$$

  - We only care about the direction of $\mathbf{w}$ so we can drop the scaling factors $(\mathbf{w}^T \mathbf{S_B}\mathbf{w})$ and $(\mathbf{w}^T \mathbf{S_W}\mathbf{w})$ which gives

$$\mathbf{S_W}\mathbf{w} = \mathbf{S_B}\mathbf{w}$$

  - $\mathbf{S_B}\mathbf{w}$ is always on the direction of $(\boldsymbol{\mu}_1 - \boldsymbol{\mu}_2)$ thus the solution is:

$$\mathbf{w} = \mathbf{S_W}^{-1}(\boldsymbol{\mu}_1 - \boldsymbol{\mu}_2)$$

Optimizing the Fisher's Linear Discriminant cost function II

- another way to solve it is to observe that the quantity:

$$J(\mathbf{w}) = \frac{\mathbf{w}^T \mathbf{S_B} \mathbf{w}}{\mathbf{w}^T \mathbf{S_W} \mathbf{w}}$$

  is known as the *generalized Rayleigh quotient*
- Then the maximizer of $J(\mathbf{w})$ is given by the generalized eigen problem :

$$\mathbf{S_B} \mathbf{w} = \lambda \mathbf{S_W} \mathbf{w}$$

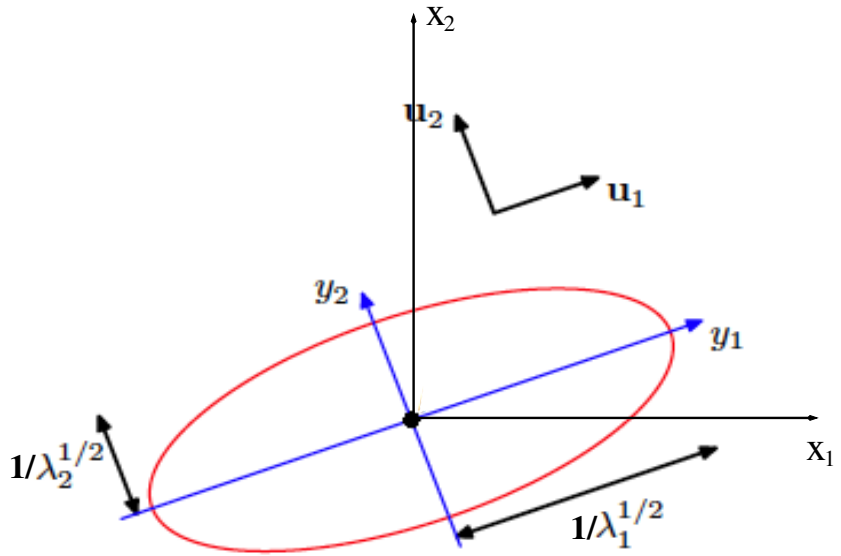
  and we need the maximum eigen vector,
- it can be shown that $\mathbf{w} = \mathbf{S_W}^{-1}(\boldsymbol{\mu}_1 - \boldsymbol{\mu}_2)$.
- In both ways of solving for $\mathbf{w}$ we are still missing w_0.

Fisher's Linear Discriminant, how to choose w_0

- Either you are ready to make some assumptions about the distribution governing your data, in which case you fall back to one of the decision rules given by equations 7,8.
- Either you choose that w_0 which maximizes separation on the training set.

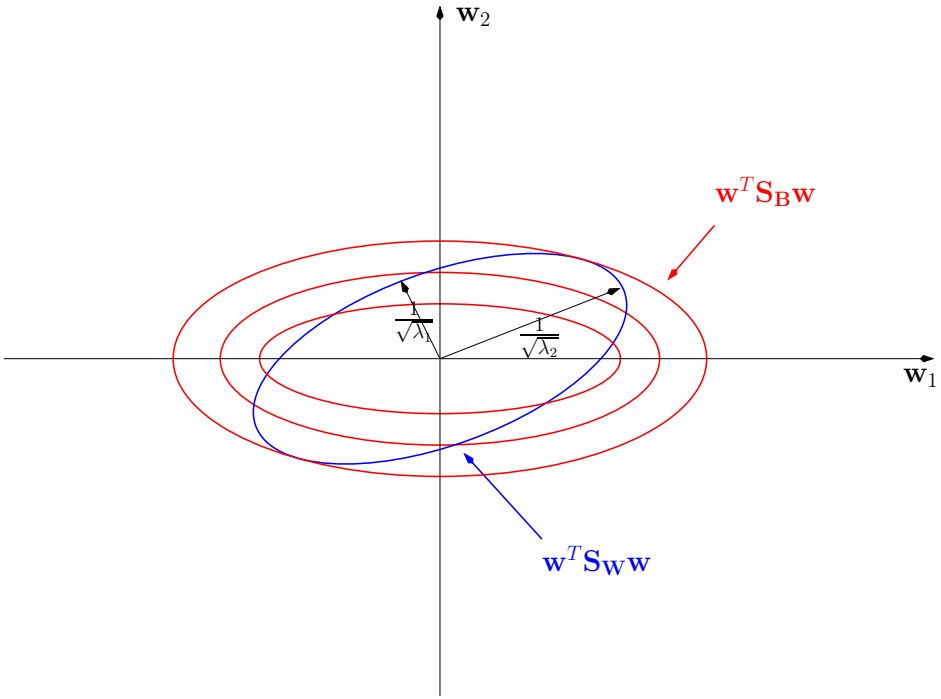
A few words on the Rayleigh quotient I

- The quadratic quantity $\mathbf{x}^T\mathbf{W}\mathbf{x}$,
- when $\mathbf{W} = \mathbf{U}\mathbf{\Lambda}\mathbf{U}^T \geq 0$, $\mathbf{U} = [\mathbf{u}_1, \mathbf{u}_2, \ldots, \mathbf{u}_d]$ ($\mathbf{u}_i$ column of $\mathbf{U}$)
- is a hyperellipsoid whose direction and length are given by the eigen vectors, $\mathbf{u}_i$, and the eigen values, λ_i, of $\mathbf{W}$ respectively.



A few words on the Rayleigh quotient II

- Thus in the Rayleigh quotient we have two kinds of hyperellipsoids defined by $\mathbf{w}^T\mathbf{S_B}\mathbf{w}$ and $\mathbf{w}^T\mathbf{S_W}\mathbf{w}$



A few words on the Rayleigh quotient III

- Note that:

$$\frac{\mathbf{w}^T\mathbf{S_B}\mathbf{w}}{\mathbf{w}^T\mathbf{S_W}\mathbf{w}} = \frac{\mathbf{w}'^T\mathbf{S_B}\mathbf{w}'}{\mathbf{w}'^T\mathbf{S_W}\mathbf{w}'}, \forall c, \mathbf{w}' = c\mathbf{w}$$

- thus to have a unique solution for $\mathbf{w}$ we need to add a constrain: $\mathbf{w}^T\mathbf{S_W}\mathbf{w} = K$
- and then maximizing the Rayleigh quotient is equivalent to:

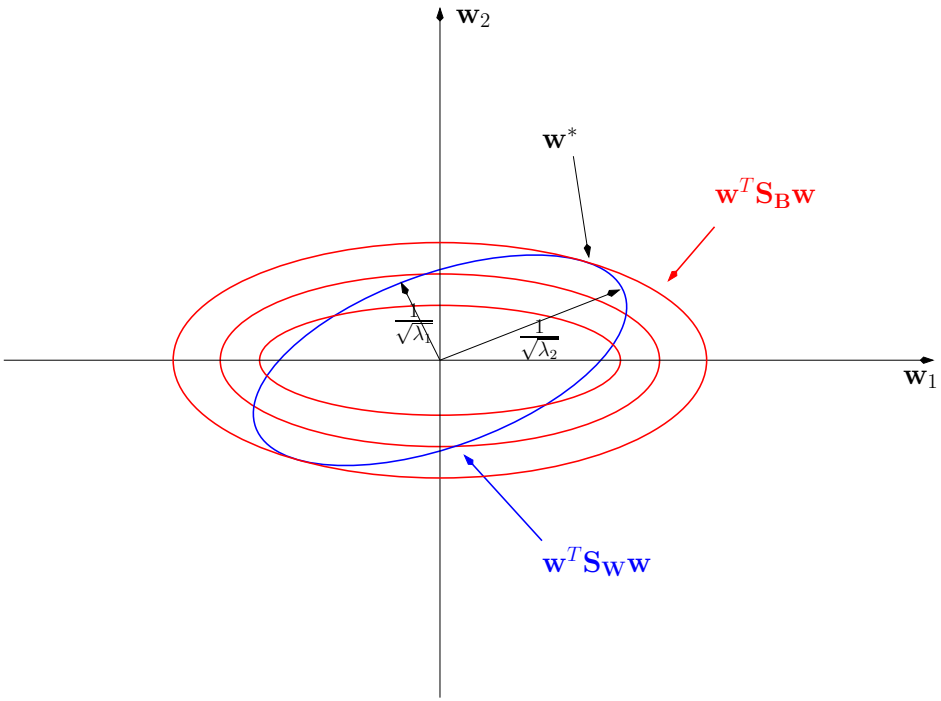
$$\max_{\mathbf{w}} \mathbf{w}^T\mathbf{S_B}\mathbf{w} \text{ subject to } \mathbf{w}^T\mathbf{S_W}\mathbf{w} = K \tag{10}$$

such equations (optimizations under constrains) are solved using the Langrange multipliers. More on the SVM course.

Rayleigh quotient visualization

- the optimal $\mathbf{w}^*$ solution of

$$\max_{\mathbf{w}} \mathbf{w}^T \mathbf{S_B} \mathbf{w} \text{ subject to } \mathbf{w}^T \mathbf{S_W} \mathbf{w} = K \tag{11}$$



is where the outer red circle, $\mathbf{w}^T \mathbf{S_B} \mathbf{w}$, touches the constrain, blue circle, $\mathbf{w}^T \mathbf{S_W} \mathbf{w}$

Perceptron

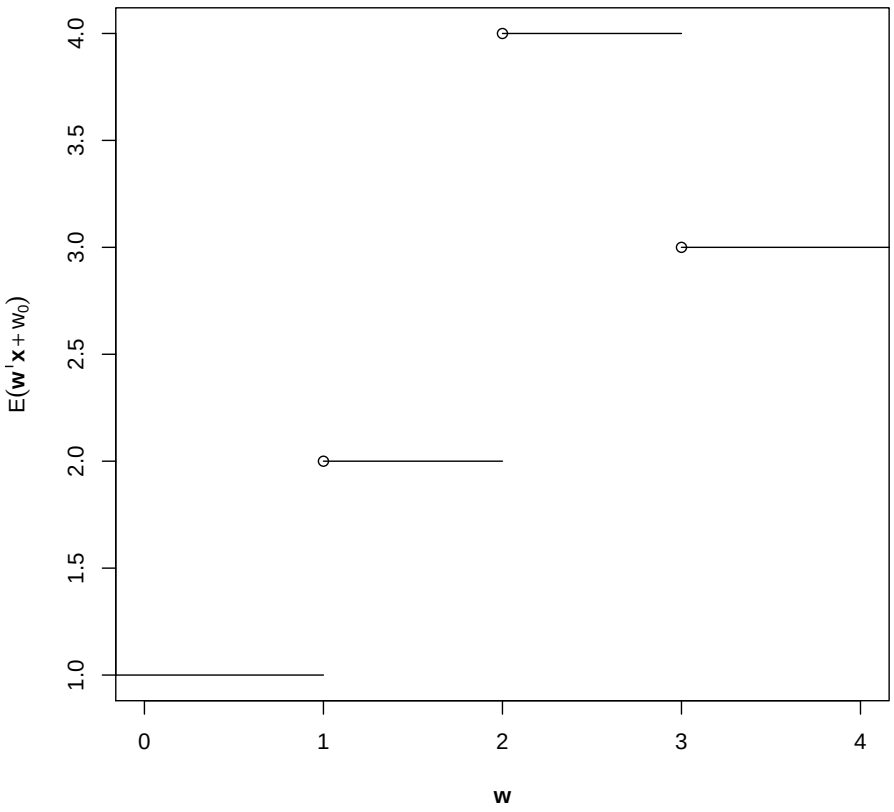
Perceptron

- Linear discriminants based on distributional assumptions
  - follow a probabilistic/generative approach estimating $p(\mathbf{x}, y)$
  - using the likelihood as a cost function to estimate the necessary parameters
- Fisher's linear discriminants
  - follow a geometric approach
  - which leads to the use of the Generalized Rayleigh quotient as the cost function
- Perceptron tries to directly optimize the *missclassification error* (or more precisely a proxy of it).

Perceptron

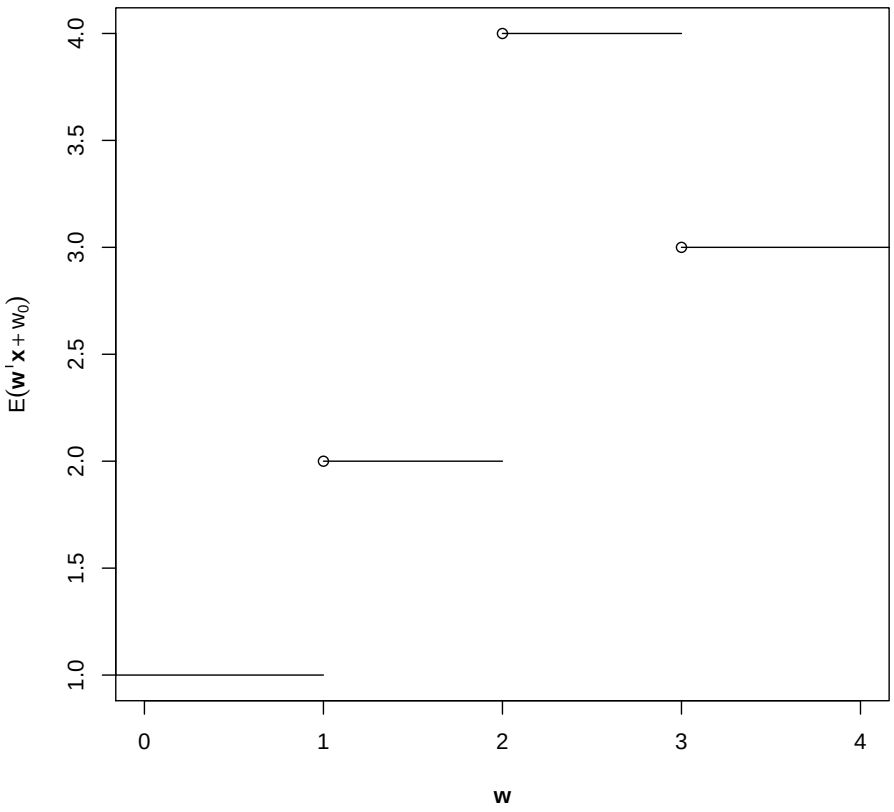
- The Perceptron algorithm provides yet another way to estimate the $\mathbf{w}$, w_0, of the $g(\mathbf{x}) = \mathbf{w}^T\mathbf{x} + w_0$ function.
- Applicable to two class problems, class encoding $c_1 = 1, c_2 = -1$
- The classification rule it uses is based on $sign(g(\mathbf{x}))$:
  - If $sign(g(\mathbf{x})) = 1$ then c_1
  - If $sign(g(\mathbf{x})) = -1$ then c_2
- Perceptron will find a set of weights *only* if the problem is linearly seperable.

Perceptron Cost function



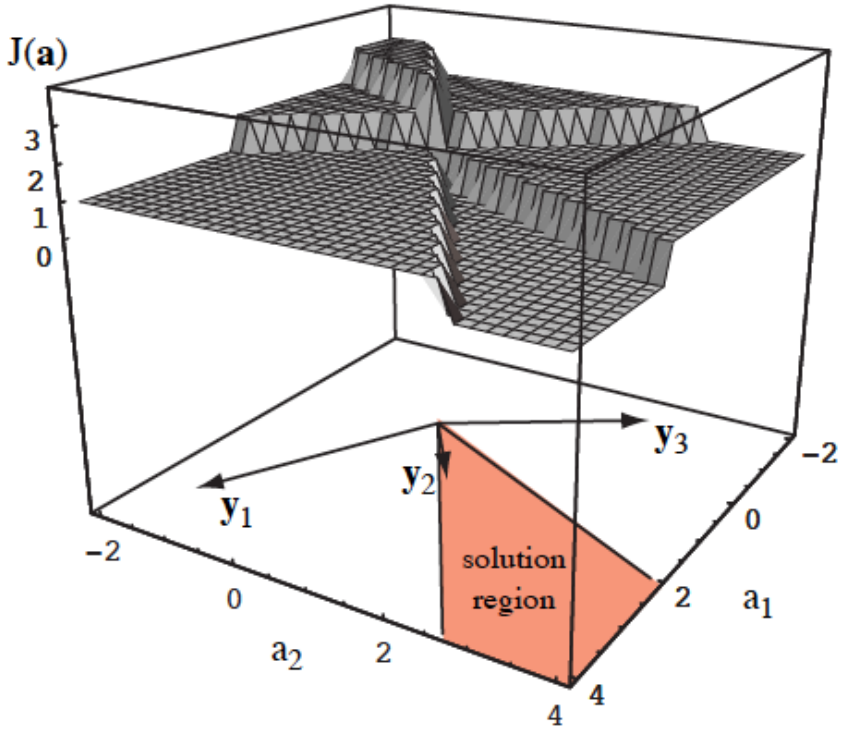
- Missclassification error $E =$ number of instances wrongly classified.
- $\arg\min_{\mathbf{w},w_0} E(\mathbf{w}^T\mathbf{x} + w_0)$
- How would you optimize it?

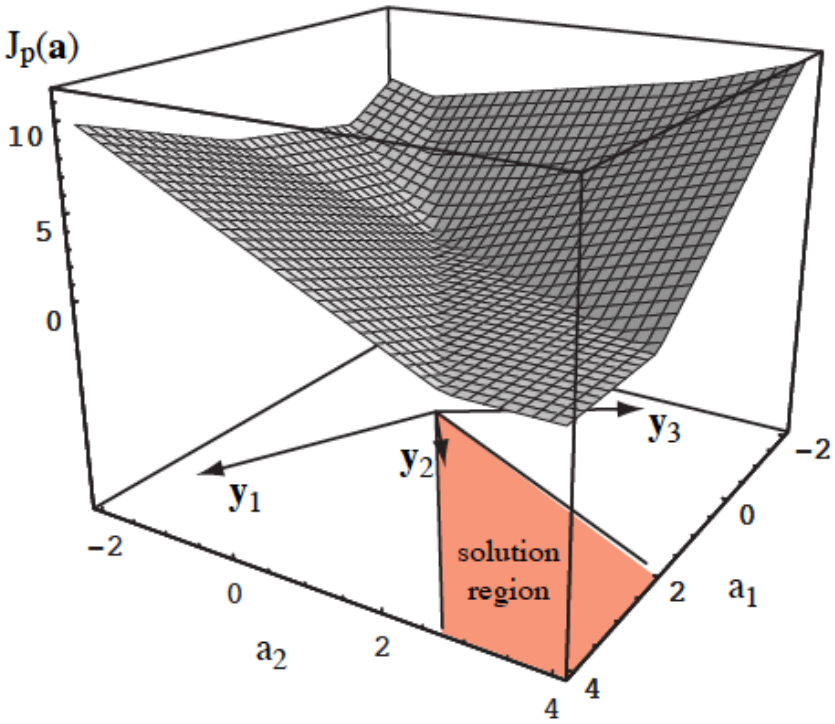
Perceptron Cost function



- Missclassification error $E =$ number of instances wrongly classified.
- $\arg\min_{\mathbf{w},w_0} E(\mathbf{w}^T\mathbf{x} + w_0)$
- How would you optimize it?
- $E(\mathbf{w}^T\mathbf{x} + w_0)$ is piecewise constant with respect to $\mathbf{w}, w_0$
- What is the gradient of this function??? zero almost everywhere...
- Rather problematic for gradient descent...

Examples of Cost functions, [4]





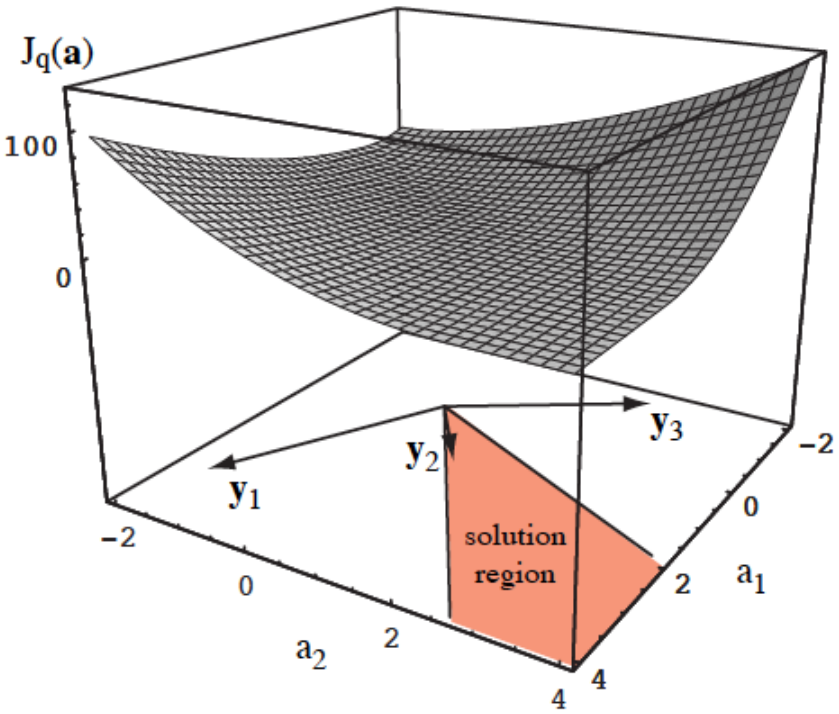
Of wrongly classified instances
Piecewise constant

$-\sum_{\mathbf{x}_i \in M}(\mathbf{w}^T\mathbf{x}_i + w_0)y_i$
Piecewise linear

y_1, y_2, y_3 training samples; a_1, a_2 model parameters; $J(a)$ cost function;

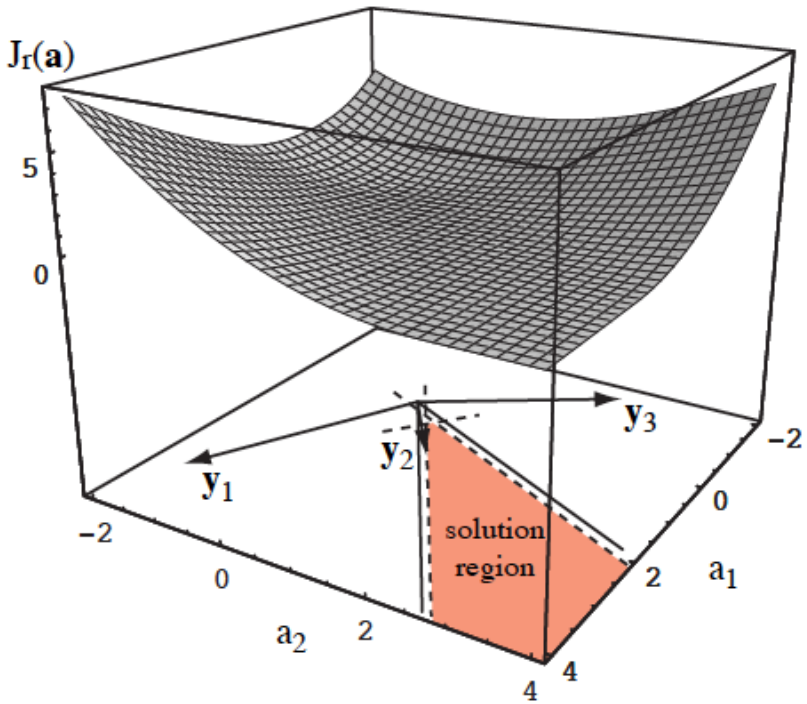
M set of missclassified instances ([4], fig 5.11

Examples of Cost functions



$\sum_{\mathbf{x}_i \in M} (\mathbf{w}^T \mathbf{x}_i + w_0)^2$

Squared Error



$\sum_{\mathbf{x}_i \in M} \frac{((\mathbf{w}^T \mathbf{x}_i + w_0) - b)^2}{||\mathbf{x}_i||^2}$

Squared Error with margin

Perceptron Cost function

- To avoid the problem with piecewise constant error functions...
- ...perceptron uses a proxy for the number of missclassification
- What is a condition that holds for $\mathbf{w}^T\mathbf{x}_i + w_0$ and y_i when the instance $\mathbf{x}_i$ is classified correctly???

Perceptron Cost function

- To avoid the problem with piecewise constant error functions...
- ...perceptron uses a proxy for the number of missclassification
- What is a condition that holds for $\mathbf{w}^T\mathbf{x}_i + w_0$ and y_i when the instance $\mathbf{x}_i$ is classified correctly???

$$(\mathbf{w}^T\mathbf{x_i} + w_0)y_i > 0$$

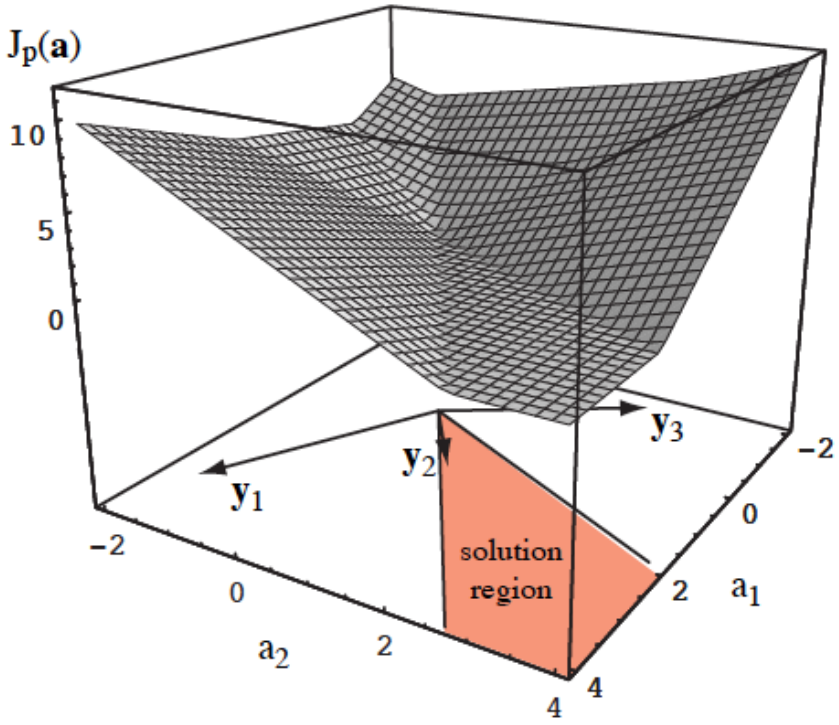
- the cost function of perceptron:
  - assigns zero error to correctly classified instances
  - tries to minimise $-(\mathbf{w}^T\mathbf{x} + w_0)y$ over the missclassified instances.

Perceptron Cost function

- Thus its cost function is given by:

$$E(\mathbf{w}, w_0) = -\sum_{\mathbf{x}_i \in M} (\mathbf{w}^T \mathbf{x}_i + w_0) y_i$$

  where M is the set of missclassified instances.
- which as we said is piecewise linear.
- $E(\mathbf{w}, w_0)$ is proportional to the sum of the distances of the missclassified examples to the decision boundary.
- There is an infinity of solutions orange area of the error graph (if of course the problem is linearly seperable).
- We prefer to have unique well defined solutions.



Optimizing the Perceptron Cost function

- We will minimize the perceptron cost function with gradient descent
- For convenience we will set:

$$\begin{aligned} \mathbf{x}^T &:= (\mathbf{x}^T, 1) \\ \mathbf{w}^T &:= (\mathbf{w}^T, w_0) \end{aligned}$$

  which simplifies:

$$\mathbf{w}^T\mathbf{x} + w_0 \text{ to } \mathbf{w}^T\mathbf{x}$$

- The gradient of

$$E(\mathbf{w}) = -\sum_{\mathbf{x}_i \in M} (\mathbf{w}^T\mathbf{x}_i) y_i$$

  is...

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$$\nabla E(\mathbf{w}) = -\sum_{\mathbf{x}_i \in M} \mathbf{x}_i y_i$$

Optimizing the Perceptron Cost function

- How can we write

$$
\begin{aligned}
E(\mathbf{w}) &= -\sum_{\mathbf{x}_i \in M} (\mathbf{w}^T \mathbf{x}_i) y_i \\
\nabla E(\mathbf{w}) &= -\sum_{\mathbf{x}_i \in M} \mathbf{x}_i y_i
\end{aligned}
$$

in matrix form?

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\end{aligned}
$$

  in matrix form?

$$
\begin{aligned}
E(\mathbf{w}) &= -[\mathbf{X}\mathbf{w}]^{\mathsf{T}} (\mathbf{y} - \mathsf{sign}(\mathbf{X}\mathbf{w})) \\
\nabla E(\mathbf{w}) &= -\mathbf{X}^{\mathsf{T}} (\mathbf{y} - \mathsf{sign}(\mathbf{X}\mathbf{w}))
\end{aligned}
$$

- You should think in terms of matrix and vector operations for computational efficiency.
- Always write your optimization problems and your algorithms in those terms.

Opitimizing the Perceptron Cost function

- Perceptron gradient descent comes into flavors:
  - *batch* the weight vector $\mathbf{w}$ is updated after seeing all the training instances, each such cycle is usually called an *epoch*
  - *single-sample* the weight vector is updated each time we see one instance (can be used to derive an on-line variant of perceptron).

Batch perceptron training

$\eta(.)$ {learning rate, usually a function of the iteration step}
θ {termination criterion}
$k \leftarrow 0$ {iteration step}
$\mathbf{w}^{(k)} \leftarrow random()$
repeat
  $k \leftarrow k + 1$
  $\mathbf{w}^{(k)} \leftarrow \mathbf{w}^{k-1} - \eta(k)\boldsymbol{\nabla} E(\mathbf{w})|_{\mathbf{w}^{(k-1)}} = \mathbf{w}^{(k-1)} + \eta(k) \sum_{\mathbf{x}_i \in M} \mathbf{x}_i y_i$
until $||\mathbf{w}^{(k)} - \mathbf{w}^{(k-1)}||_2 = ||\eta(k) \sum_{\mathbf{x}_i \in M} \mathbf{x}_i y_i||_2 \leq \boldsymbol{\theta}$

Single sample perceptron training

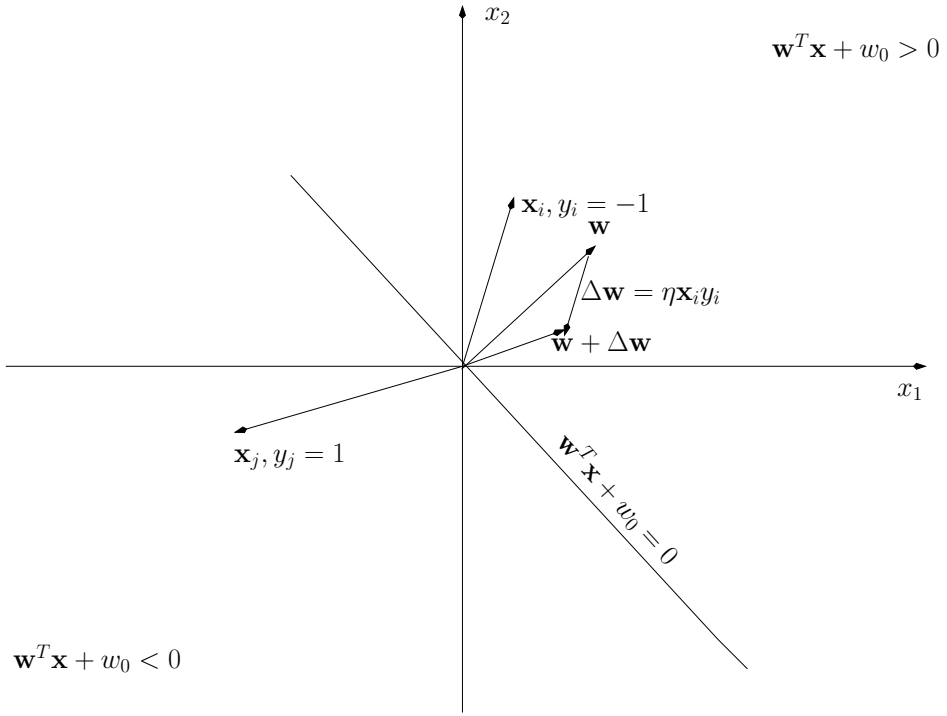
$\eta(.)$ {learning rate, usually a function of the iteration step}
$\boldsymbol{\theta}$ {termination criterion}
$k \leftarrow 0$ {iteration step}
$\mathbf{w} \leftarrow random()$
repeat
  for all $\mathbf{x}_i$ **do**
    if $(\mathbf{w}^T \mathbf{x}_i y_i < 0)$ **then**
      $\mathbf{w} \leftarrow \mathbf{w} + \eta(k)\mathbf{x}_i y_i$
    end if
  end for
until all $\mathbf{x}_i$ correctly classified

Perceptron, weight updating

- Intuitevely when there is a missclassification, i.e. $y_i(\mathbf{w}^{(k)}\mathbf{x}_i) < 0$, of an instance $\mathbf{x_i}$ the weight vector

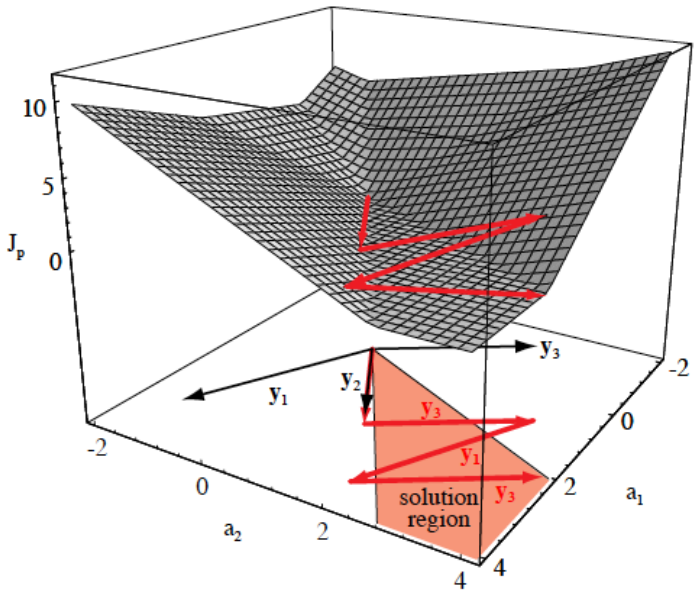
Perceptron, weight updating

- Intuitevely when there is a missclassification, i.e. $y_i(\mathbf{w}^{(k)}\mathbf{x}_i) < 0$, of an instance $\mathbf{x_i}$ the weight vector $\mathbf{w}^{(k)}$ moves on a direction determined by $y_i\mathbf{x}_i$.
- Whenever there is a misclassification the hyperplane moves
  - to the direction of the missclassified example, if it is a positive one
  - oposite of the direction of the missclassified, if it is a negative one



Perceptron, weight updating

- Concecutive weight vectors $\mathbf{w}^{(k)}$ in perceptron update.
- Note how in the second update the weight vector moves further from the solution region.



Controlling the learning rate in (gradient) descent methods

- $\eta(.)$ controls the rate of convergence
  - too small a $\eta(.)$ $\Rightarrow$ too slow convergence
  - too large a $\eta(.)$ $\Rightarrow$ overshooting or even divergence
- Different strategies to control $\eta(.)$ [3, 1]
  - *Heuristic*: start with large η and decrease it with the number of iterations
  - *Exact Line search*: exactly compute η that results in the largest error reduction.
  - *Inexact line search*: compute some η that gives a good error reduction.
    - Backtracking line search

Exact line search

- Main idea: define a one variable, η, optimization problem and solve it at each iteration step of the descent method.

Exact line search

- Let $\Delta\mathbf{w}^{(k)}$ be a descent direction at step k; for example in gradient descent

$$\Delta\mathbf{w}^{(k)} = -\nabla E(\mathbf{w})|_{\mathbf{w}^{(k-1)}}$$

- Define the single variable function:

$$h(\eta) \quad = \quad E(\mathbf{w}^{(k-1)} + \eta\Delta\mathbf{w}^{(k)})$$

to simplify notation we drop the iteration index:

$$h(\eta) = E(\mathbf{w} + \eta\Delta\mathbf{w})$$

- and solve the optimization problem:

$$\eta^{(k)} = \arg\min_{\eta} h(\eta) \tag{12}$$

- Exact line search is used when the cost of solving eq 12 is small compared to cost of establishing the direction [3].

Backtracking line search

- Backtracking line search is an inexact line search in which we look for a good step length η and not necessarily the one that maximily reduces our error function.

Backtracking line search

- $E(\mathbf{w} + \eta \Delta \mathbf{w}) = h(\eta) \sim E(\mathbf{w}) + \eta \nabla E(\mathbf{w})^T \Delta \mathbf{w}$
- For a $\Delta \mathbf{w}$ small enough and given that it is a descent direction then $\nabla E(\mathbf{w})^T \Delta \mathbf{w}$

Backtracking line search

- $E(\mathbf{w} + \eta\Delta\mathbf{w}) = h(\eta) \sim E(\mathbf{w}) + \eta\nabla E(\mathbf{w})^T\Delta\mathbf{w}$
- For a $\Delta\mathbf{w}$ small enough and given that it is a descent direction then $\nabla E(\mathbf{w})^T\Delta\mathbf{w} < 0$
- and given $a \in (0, 0.5)$ we have:

$$\begin{aligned} E(\mathbf{w} + \eta\Delta\mathbf{w}) &= h(\eta) \\ &\sim E(\mathbf{w}) + \eta\nabla E(\mathbf{w})^T\Delta\mathbf{w} \\ &< E(\mathbf{w}) + a\eta\nabla E(\mathbf{w})^T\Delta\mathbf{w} \end{aligned}$$

- A learning rate η is acceptable as long as:

$$E(\mathbf{w} + \eta\Delta\mathbf{w}) \leq E(\mathbf{w}) + a\eta\nabla E(\mathbf{w})^T\Delta\mathbf{w}$$

  otherwise we reduce it to:

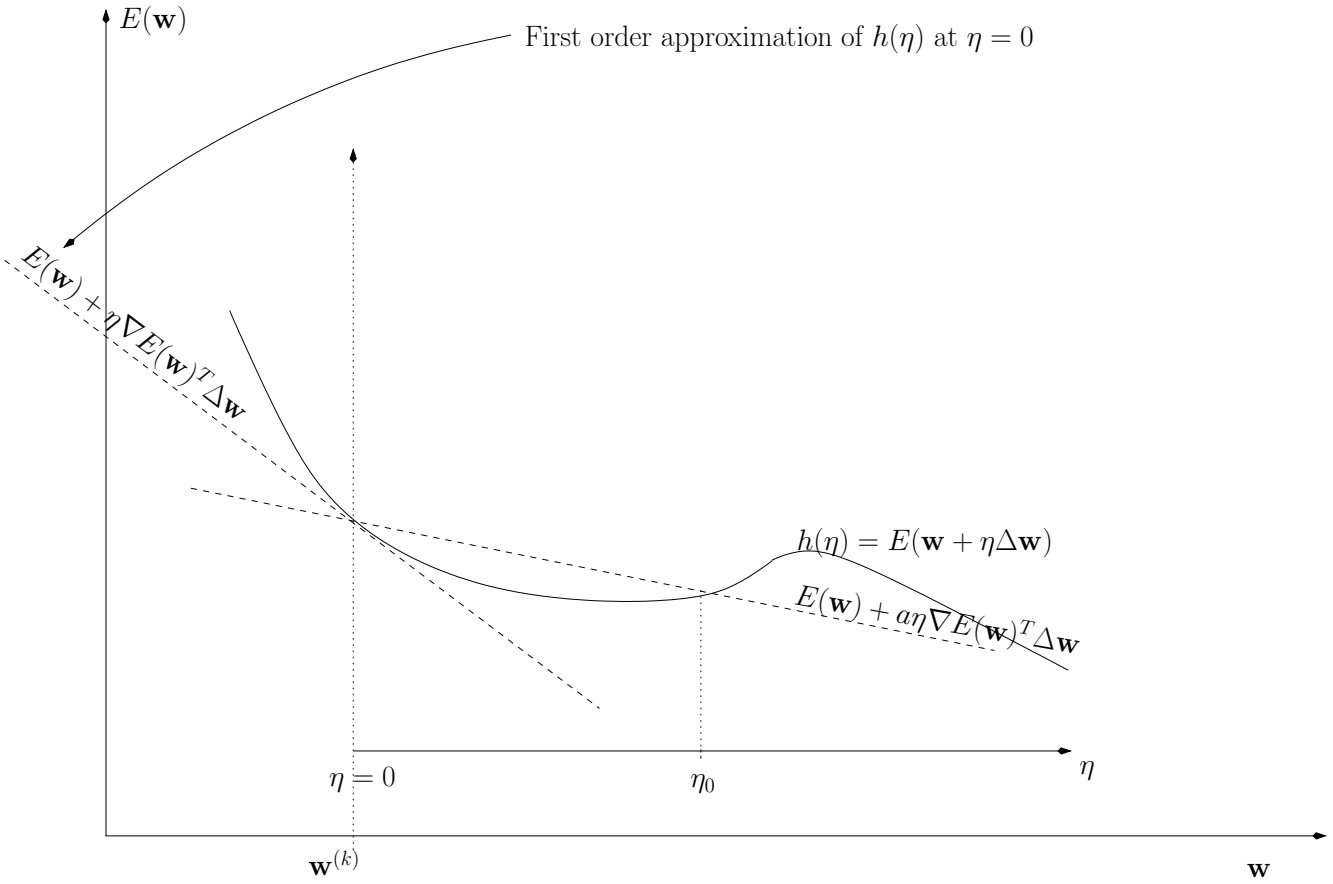
$$\eta \leftarrow \beta\eta$$

  where $\beta \in (0, 1)$

Backtracking line search, Algorithm

$\Delta\mathbf{w}$ {Descent direction, determined for example at each gradient descent step.}
$a \in (0, 0.5)$
$\beta \in (0, 1)$
$\eta := 1$ {Initial learning rate.}
while $E(\mathbf{w} + \eta\Delta\mathbf{w}) > E(\mathbf{w}) + a\eta\nabla E(\mathbf{w})^T\Delta\mathbf{w}$ **do**
  $\eta \leftarrow \beta\eta$
end while

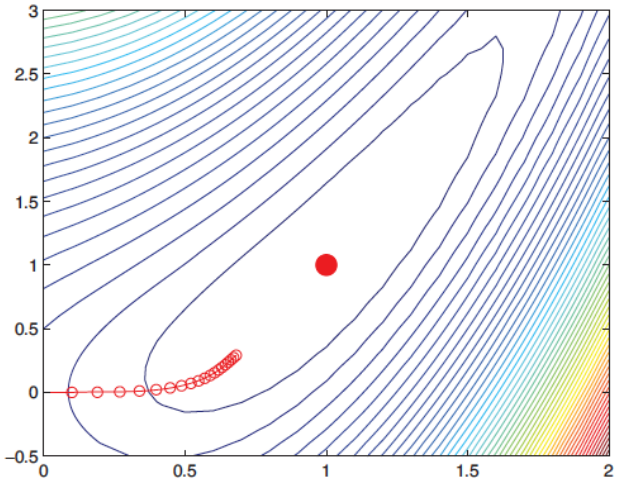
Backtracking line search [3, 1]



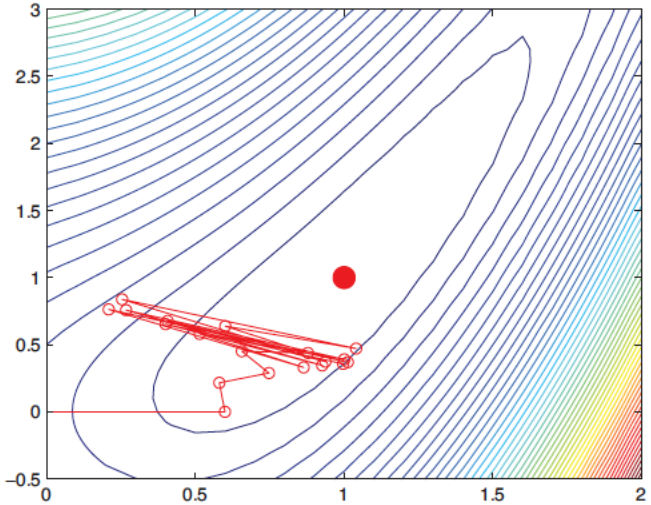
- a is the % of decrease of $E(\mathbf{w} + \eta \Delta \mathbf{w})$ that we accept, with respect to that predicted by its linear approximation $E(\mathbf{w}) + \eta \nabla E(\mathbf{w}) \Delta \mathbf{w}$

Gradient descent examples, [5]

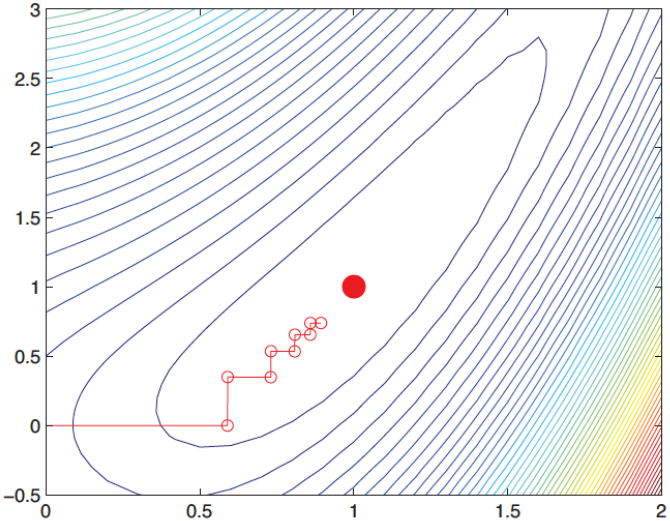
$\eta = 0.1$



$\eta = 0.6$



Exact line search



Logistic regression

Logistic Regression

- In LD with distributional assumptions we used the joint distribution $p(\mathbf{x}, y)$ an approach that is called *generative*.
- Perceptron belongs to the *discriminant function* approaches, which learn directly $g(\mathbf{y}) = y$.
- In logistic regression we will directly estimate $p(y|\mathbf{x})$ an approach that is called *discriminative*.
- Before going to logistic regression we should review the Bernoulli distribution

The Bernoulli distribution

- Consider a random variable y that takes values $\in \{0, 1\}$
- For example y could represent the result of flipping a coin
  - $y = 1$ represents heads
  - $y = 0$ represents tails

  our coin does not need to be fair
- we will denote with μ the probability of $y = 1$, thus:

$$p(y = 1|\mu) = \mu$$

  obviously:

$$p(y = 0|\mu) = 1 - p(y = 1|\mu) = 1 - \mu$$

- obviously:
  - $0 \leq \mu \leq 1$, and
  - $p(y = 0|\mu) = 1 - \mu$

- The probability distribution of y is given by

$$y \sim \text{Bern}(y|\mu) = \mu^y (1-\mu)^{(1-y)} \tag{13}$$

  known as the *Bernoulli distribution*
- Use equation 13 to get the probability of $y = 1$ or $y = 0$. What does it give?
- Its mean and variance are given by

$$\begin{aligned} \mathbb{E}[y] &= \mu \\ \text{var}[y] &= \mu(1-\mu) \end{aligned}$$

Likelihood function of a Bernoulli distribution

- We have a dataset D of n samples drawn i.i.d from that distribution:

$$D = \{y_1, y_2, \ldots y_n\}$$

- Then the *likelihood* of that dataset is:

$$p(D|\mu) =$$

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- As we said the likelihood is a *function* of the parameters of your distribution/model, what are these parameters here?

Likelihood function of a Bernoulli distribution

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- As we said the likelihood is a *function* of the parameters of your distribution/model, what are these parameters here? μ
- Again as already discussed, different notations can be used for the likelihood function, e.g.

$$l(D; \boldsymbol{\theta}),\ l(\boldsymbol{\theta})$$

$\boldsymbol{\theta}$ the parameters of your distribuition/model (here $\boldsymbol{\theta} = \{\mu\}$).

Log Likelihood function and negative log likelihood

- Given a dataset $D = \{y_1, ..., y_n\}$ of i.i.d samples the likelihood function is always a product:

$$p(D|\boldsymbol{\theta}) = \prod_{i=1}^{n} p(y_i|\theta)$$

- As we already saw it is very convinient to take the log of the likelihood, which gives:

$$\ln p(D|\boldsymbol{\theta}) = \ln \prod_{i=1}^{n} p(y_i|\theta) = \sum_{i=1}^{n} \ln p(y_i|\theta)$$

- very often we will be using the *negative log likelihood*, i.e.

$$-\ln p(D|\boldsymbol{\theta})$$

(similar to error, want to make it small)

Log Likelihood function of the Bernoulli

$$
\begin{aligned}
\ln \prod_{i=1}^{n} \mu_i^{y} (1-\mu)^{(1-y_i)} &= \sum_{i=1}^{n} \ln \mu^{y_i} (1-\mu)^{(1-y_i)} \\
&= \sum_{i=1}^{n} \ln \mu^{y_i} + \sum_{i=1}^{n} (1-\mu)^{(1-y_i)} \\
&= \sum_{i=1}^{n} y_i \ln \mu + \sum_{i=1}^{n} (1-y_i) \ln(1-\mu) \quad (14)
\end{aligned}
$$

Maximum Likelihood Estimation

- In statistics we do parameter estimation by *maximizing the likelihood*
- Maximum Likelihood Estimation (MLE) is general statistical estimation approach to estimate the parameters of a distribution/model given a dataset
- It looks for the parameters that maximize the dataset likelihood, i.e. in the case of the Bernoulli distribution:

$$\max_{\mu} p(D|\mu)$$

- We can choose to maximise the likelihood or the log likelihood, but we can also (equivalently) minimize the negative log likelihood.
- How to get the Maximum Likelihood Estimate of μ ?

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- We can choose to maximise the likelihood or the log likelihood, but we can also (equivalently) minimize the negative log likelihood.
- How to get the Maximum Likelihood Estimate of μ ? get the derivative of the likelihood, set it to zero and solve for the parameters μ

Maximum Log Likelihood Estimation for the Bernoulli

- find μ that maximises the log likelihood of the Bernoulli eq 14
- If you do it right you should get

$$\mu = \frac{1}{n} \sum_{i=1}^{n} y_i$$

For more on the Bernoulli distribution check [2] second chapter.

Back to Logistic regression

Logistic Regression, I

- We are given:
  - two classes C_1 and C_2 with respective class labels 1 and 0
  - $\mathbf{X}$ the matrix with lines the training instances, and
  - $\mathbf{y}$ the respective vector of class labels.

Logistic Regression, II

- The class y of some instance $\mathbf{x}$ follows a Bernoulli distribution.

  $$y \sim \text{Ber}(y|\mu(\mathbf{x})) = [p(y = C_1|\mu(\mathbf{x}))]^y [1 - p(y = C_1|\mu(\mathbf{x}))]^{1-y}$$

  now the "μ" is different for each instance $\mathbf{x}$
- Logistic regression will learn to estimate this conditional distribution, i.e. $P(y_i|\mu(\mathbf{x}_i))$, in particular it will learn to estimate $p(y = C_1|\mathbf{x})$ in the case of a two class problem.

Logistic Regression Likelihood Function

- The likelihood of the vector of observed class labels $\mathbf{y}$ is given by:

Logistic Regression Likelihood Function

- The likelihood of the vector of observed class labels $\mathbf{y}$ is given by:

$$
\begin{aligned}
p(\mathbf{y}|\mathbf{X}) &= \prod_{i=1}^{N} p(y_i|\mathbf{x}_i) \\
&= \prod_{i=1}^{N} [p(y_i = C_1|\mathbf{x}_i)]^{y_i} [1 - p(y_i = C_1|\mathbf{x}_i)]^{1-y_i}
\end{aligned}
$$

- Thus logistic regression will maximise the likelihood (in a similar manner to the simple case of μ) by learning to estimate $P(y|\mathbf{x})$

Logistic Regression, modelling $P(y|\mathbf{x})$, I

- As we said the label y of some instance $\boldsymbol{x}$:

$$y \sim \text{Ber}(y|\mu(\mathbf{x})), \text{ governed by } \mu(\boldsymbol{x}) := p(y = C_1|\mathbf{x})$$

- to simplify notation let:

$$p(C_i|\mathbf{x}) := p(y = C_i|\mathbf{x})$$

- where we compute $p(C_1|\mathbf{x})$ as:

$$p(C_1|\mathbf{x})$$

Logistic Regression, modelling $P(y|\mathbf{x})$, I

- As we said the label y of some instance $\boldsymbol{x}$:

$$y \sim \text{Ber}(y|\mu(\mathbf{x})), \text{ governed by } \mu(\boldsymbol{x}) := p(y = C_1|\mathbf{x})$$

- to simplify notation let:

$$p(C_i|\mathbf{x}) := p(y = C_i|\mathbf{x})$$

- where we compute $p(C_1|\mathbf{x})$ as:

$$\begin{aligned} p(C_1|\mathbf{x}) &= \frac{p(\mathbf{x}|C_1)p(C_1)}{p(\mathbf{x}|C_1)p(C_1) + p(\mathbf{x}|C_2)p(C_2)} \\ &= \frac{1}{1 + \exp(-a)} = \sigma(a) \end{aligned} \tag{15}$$

Logistic Regression, modelling $P(y|\mathbf{x})$, II I

- $\sigma(a)$ is the *logistic sigmoid function*, used also a lot in neural network algorithms.
- if we solve for a in eq 15 we get:

$$a = \ln \frac{p(\mathbf{x}|C_1)p(C_1)}{p(\mathbf{x}|C_2)p(C_2)} = \ln \frac{p(C_1|\mathbf{x})}{p(C_2|\mathbf{x})}$$

known as the *log odds ratio* or the *logit* function is the inverse of the $\sigma(a)$ function.

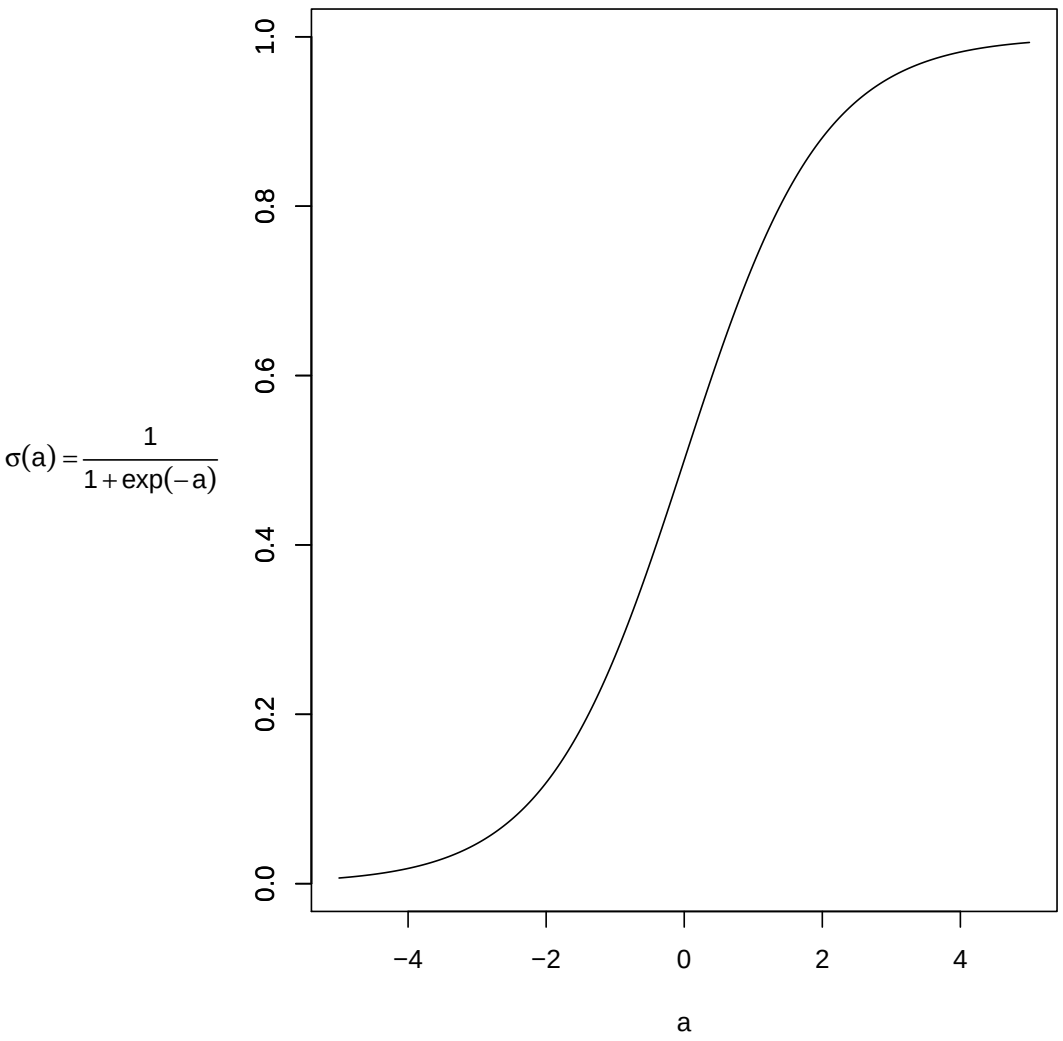
The logistic sigmoid function, $\sigma(a)$

- It squezes all the real axis in a finite interval.
- $\sigma(-a) = 1 - \sigma(a)$
- its inverse function is:

$$a = \ln \frac{\sigma(a)}{1 - \sigma(a)} = \ln \frac{p(C_1|\mathbf{x})}{p(C_2|\mathbf{x})}$$

the *logit function*, *log odds*

- $\frac{d\sigma(a)}{da} = \sigma(a)(1 - \sigma(a))$



A multiclass generalization of logistic sigmoid function

- When we have more than two classes, instead of the logistic sigmoid we use:

$$
\begin{aligned}
p(C_k|\mathbf{x}) &= \frac{p(\mathbf{x}|C_k)p(C_k)}{\sum_k p(\mathbf{x}|C_k)p(C_k)} \\
&= \frac{\exp a_k}{\sum_k \exp a_k}
\end{aligned}
$$

known as the *soft max*, $a_k = \ln p(\mathbf{x}|C_k)p(C_k)$

- What's the use of the re-writting we did?

$$
\begin{aligned}
p(C_1|\mathbf{x}) &= \frac{p(\mathbf{x}|C_1)p(C_1)}{p(\mathbf{x}|C_1)p(C_1) + p(\mathbf{x}|C_2)p(C_2)} \\
&= \frac{1}{1 + \exp(-a)} = \sigma(a)
\end{aligned}
$$

- What's the use of the re-writting we did?

$$\begin{aligned} p(C_1|\mathbf{x}) &= \frac{p(\mathbf{x}|C_1)p(C_1)}{p(\mathbf{x}|C_1)p(C_1) + p(\mathbf{x}|C_2)p(C_2)} \\ &= \frac{1}{1+\exp(-a)} = \sigma(a) \end{aligned}$$

we will make a a function of $\mathbf{x}$, and more precisely:

$$a(\mathbf{x}) = \mathbf{w}^T\mathbf{x}$$

- for more than two classes and the softmax:

$$p(C_k|\mathbf{x}) = \frac{\exp a_k}{\sum_k \exp a_k}$$

we have:

$$a_k(\mathbf{x}) = \mathbf{w}_k^T\mathbf{x}$$

Logistic regression cost function

- Its cost function is, the negative log likelihood:

$$
\begin{aligned}
E(\mathbf{w}) &= -\ln p(\mathbf{y}|\mathbf{X}) = -\ln \prod_{i=1}^{N} p(C_1|\mathbf{x}_i)^{y_i}(1-p(C_1|\mathbf{x}_i))^{1-y_i} \\
&= -\sum_{i=1}^{N} y_i \ln p(C_1|\mathbf{x}_i) - \sum_{i=1}^{N}(1-y_i)\ln(1-p(C_1|\mathbf{x}_i)) \\
&= -\sum_{i=1}^{N} y_i \ln \sigma(\mathbf{w}^T\mathbf{x}_i) - \sum_{i=1}^{N}(1-y_i)\ln(1-\sigma(\mathbf{w}^T\mathbf{x}_i))
\end{aligned}
$$

which we minimize, thus we fall under the *Maximum Likelihood Estimation* paradigm. Its gradient is:

$$
\nabla E(\mathbf{w}) = \sum_{i=1}^{N}(\sigma(\mathbf{w}^T\mathbf{x}_i) - y_i)\mathbf{x}_i = \mathbf{X}^\mathsf{T}(\sigma(\mathbf{X}\mathbf{w}) - \mathbf{y})
$$

Logistic regression cost function/cross entropy

- if we denote by

$$Q_i(C|\sigma(\mathbf{x}_i)) = [P(C_1|\sigma(\mathbf{w}^T\mathbf{x}_i)), P(C_2|\sigma(\mathbf{w}^T\mathbf{x}_i))]$$

  the learned/predicted posterior of the class for the $\mathbf{x}_i$,
- and by

$$P_i(C|x_i) = [y_i, 1 - y_i]$$

  the true posterior,
- then we can rewritte the cost function as:

$$\sum_{i=1}^{N} H(P_i, Q_i)$$

- where

$$H(P, Q) = -\sum_{z} P(z) \ln Q(z)$$

  is the *cross entropy function* measuring distributions distance.
- Thus we seek $\mathbf{w}$ that minimizes the distance of the true from the learned posterior.

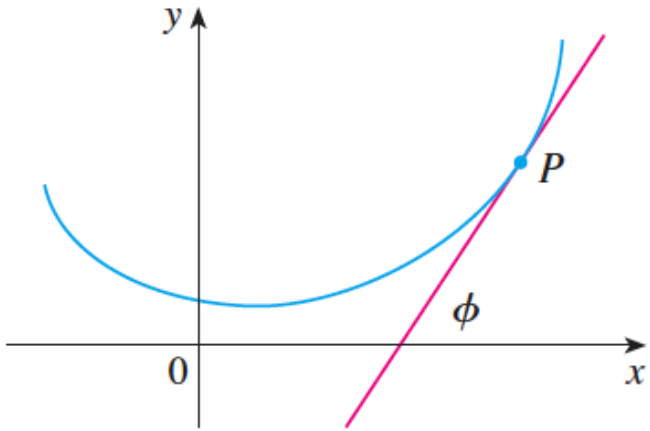
Logistic regression objective function and its relation to the objective functions of other methods

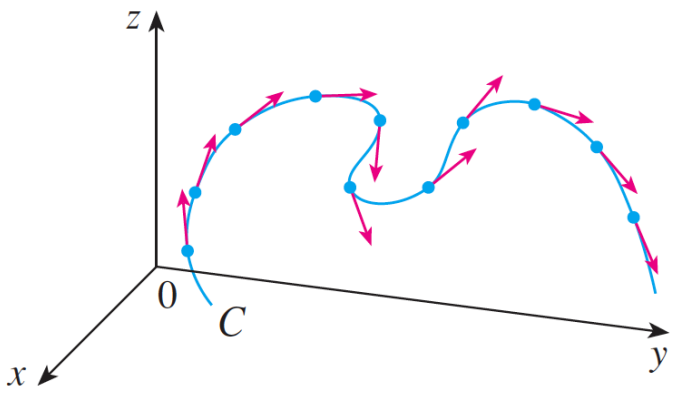
	Objective Function	Gradient
Perceptron	$\min_{\mathbf{w}} -\sum_{i=1}^{N} \mathbf{w}^{\mathsf{T}}\mathbf{x}_i(y_i - \text{sign}(\mathbf{w}^{\mathsf{T}}\mathbf{x}_i)$	$\sum_{i=1}^{N} (\text{sign}(\mathbf{w}^{\mathsf{T}}\mathbf{x}_i) - y_i)\mathbf{x}_i$
Log. R.	$\min_{\mathbf{w}} -\ln \prod_{i=1}^{N} p(C_1\|\mathbf{x}_i)^{y_i}(1 - p(C_2\|\mathbf{x}_i))^{1-y_i}$	$\sum_{i=1}^{N}(\sigma(\mathbf{w}^{\mathsf{T}}\mathbf{x}_i) - y_i)\mathbf{x}_i$
Lin. R.	$\max_{\mathbf{w}} \ln \prod_{i=1}^{N} N(y_i\|\mathbf{w}^{T}\mathbf{x}_i, \sigma^2)$	$\sum_{i=1}^{N}(\mathbf{w}^{\mathsf{T}}\mathbf{x}_i - y_i)\mathbf{x}_i$

- In perceptron the update is done for each missclassified instance
- In Log. R. the update is done according to the error of each instance
- Note the similarity of the gradient of Log.R. to that of Lin. R.
- In your implementations you should use the matrix variants of the gradients, see for example page 144 for logistic regression.

Optimizing the logistic regression cost function

- Obvious choice is the gradient descent
- Faster optimization methods take into account the *curvature* of the function





- The curvature at a point P is the rate of change of the angle ϕ with respect to the arc s, given by $|\frac{d\phi}{ds}|$, it is a measure of how fast the curve changes direction at P, [6].
- the second order derivative, Hessian, is a measure of the local curvature.

Newton's method

- Newton's method uses the curvature of the function through the Hessian
- such methods are called *second order* optimization methods

Reminder, Hessian

- For a function $f(\mathbf{x}) : R^d \to R$
- $\nabla^2 f(\mathbf{x})$ is the Hessian of f, which is the $d \times d$ matrix:

$$\nabla^2 f(\mathbf{x}) = \mathbf{H}(\mathbf{x}) = \begin{bmatrix} \frac{\partial^2 f}{\partial x_1^2} & \frac{\partial^2 f}{\partial x_1 \partial x_2} & \cdots & \frac{\partial^2 f}{\partial x_1 \partial x_d} \\ \frac{\partial^2 f}{\partial x_2 \partial x_1} & \frac{\partial^2 f}{\partial x_2^2} & \cdots & \frac{\partial^2 f}{\partial x_2 \partial x_d} \\ \vdots & \vdots & \ddots & \vdots \\ \frac{\partial^2 f}{\partial x_d \partial x_1} & \frac{\partial^2 f}{\partial x_d \partial x_2} & \cdots & \frac{\partial^2 f}{\partial x_d^2} \end{bmatrix}$$

Reminder, second order Taylor expansion

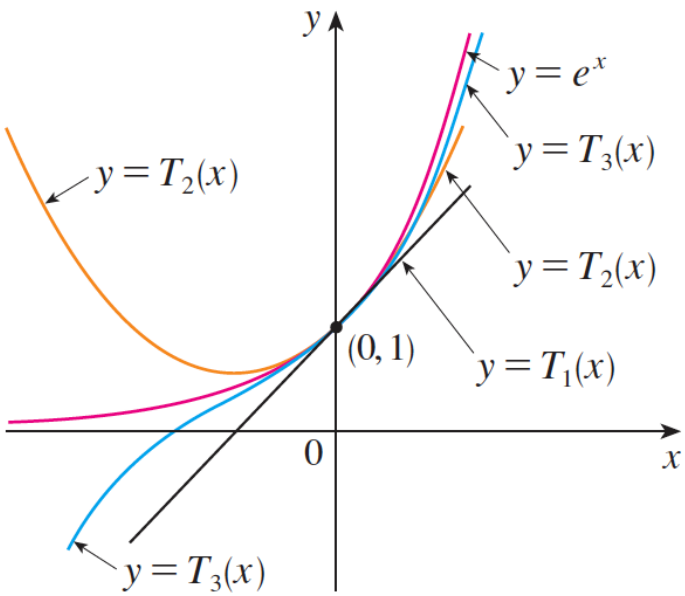
- The second order Taylor expansion of a function $f(.)$ on a point $\mathbf{z}$ around point $\mathbf{x}$, i.e. $\mathbf{z} = \mathbf{x} + \mathbf{v}$, $\mathbf{v} \to 0$, is given by

$$f(\mathbf{x} + \mathbf{v}) \simeq f(\mathbf{x}) + \nabla f(\mathbf{x})^T \mathbf{v} + \mathbf{v}^T \nabla^2 f(\mathbf{x}) \mathbf{v}$$

or

$$f(\mathbf{z}) \simeq f(\mathbf{x}) + \nabla f(\mathbf{x})^T (\mathbf{z} - \mathbf{x}) + (\mathbf{z} - \mathbf{x})^T \nabla^2 f(\mathbf{x}) (\mathbf{z} - \mathbf{x})$$

Examples of the first three order Taylor approximations of $f(x) = e^x$.



Newton's method

- Let $E(\mathbf{w})$ be a function which we want to optimize.
- Let $\mathbf{w}^{(k)}$ be the current estimate of the parameter vector.
- Then we can take its second order expansion around $\mathbf{w}^{(k)}$

$$\begin{aligned} E(\mathbf{w}) &= E(\mathbf{w}^{(k)} + \mathbf{h}) \\ &\simeq E(\mathbf{w}^{(k)}) + \nabla_{\mathbf{w}} E(\mathbf{w}^{(k)})^T \mathbf{h} + \frac{1}{2} \mathbf{h}^T \nabla^2_{\mathbf{w}} E(\mathbf{w}^{(k)}) \mathbf{h} \end{aligned}$$

- and search for that direction $\mathbf{h}$ that will optimize $E(\mathbf{w})$:

$$\max_{\mathbf{w}} / \min_{\mathbf{w}} E(\mathbf{w}) = \max_{\mathbf{h}} / \min_{\mathbf{h}} E(\mathbf{w}^{(k)} + \mathbf{h})$$

- In fact whether we $\min$ or $\max$ depends on the function $E(\mathbf{w})$

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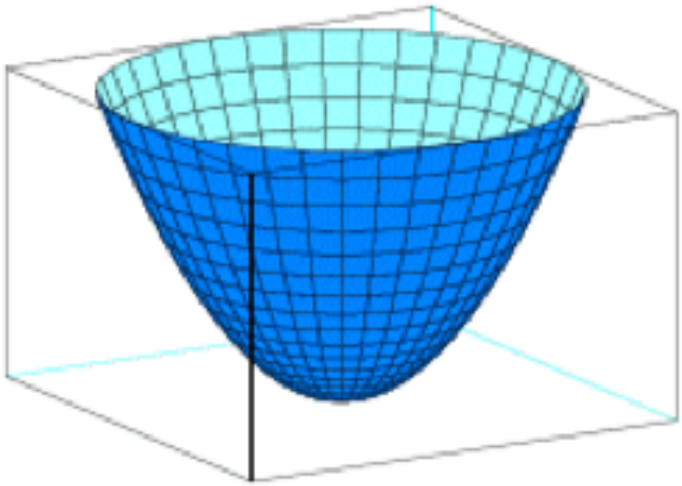
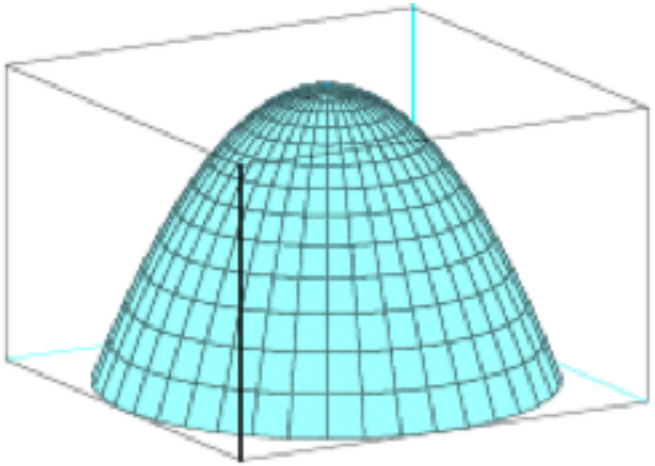
- and search for that direction $\mathbf{h}$ that will optimize $E(\mathbf{w})$:

$$\max_{\mathbf{w}} / \min_{\mathbf{w}} E(\mathbf{w}) = \max_{\mathbf{h}} / \min_{\mathbf{h}} E(\mathbf{w}^{(k)} + \mathbf{h})$$

- In fact whether we $\min$ or $\max$ depends on the function $E(\mathbf{w})$
  - If $\mathbf{x}^T \nabla^2 E(\mathbf{w})\mathbf{x} \geq 0, \forall x$ then $E(\mathbf{w})$ is convex and has a unique minimum
  - If $-\mathbf{x}^T \nabla^2 E(\mathbf{w})\mathbf{x} \geq 0, \forall x$ then $E(\mathbf{w})$ is concave and has a unique maximum

Newton's method (Convex vs Concave)





convex $f(\mathbf{w}) = \mathbf{w}^T\mathbf{w}$

$$\nabla f(\mathbf{w}) = \mathbf{w}$$

$$\nabla^2 f(\mathbf{w}) = \mathbf{I}$$

concave $f(\mathbf{w}) = -\mathbf{w}^T\mathbf{w}$

$$\nabla f(\mathbf{w}) = -\mathbf{w}$$

$$\nabla^2 f(\mathbf{w}) = -\mathbf{I}$$

Newton's method

- The optimizing direction $\mathbf{h}$ is given by:

Newton's method

- The optimizing direction $\mathbf{h}$ is given by:

$$\begin{aligned}\nabla_{\mathbf{h}} E(\mathbf{w}^{(k)} + \mathbf{h}) = \mathbf{0} \quad &\Rightarrow \quad \nabla_{\mathbf{w}} E(\mathbf{w}^{(k)}) + \nabla^2_{\mathbf{w}^{(k)}} E(\mathbf{w}^{(k)})\mathbf{h} = \mathbf{0} \\ &\Rightarrow \quad \mathbf{h} = -(\nabla^2_{\mathbf{w}} E(\mathbf{w}^{(k)}))^{-1} \nabla_{\mathbf{w}} E(\mathbf{w}^{(k)}) \\ &\qquad\quad = -(\mathbf{H}(\mathbf{w}^{(k)}))^{-1} \nabla_{\mathbf{w}} E(\mathbf{w}^{(k)})\end{aligned}$$

- thus:

$$\mathbf{w}^{(k+1)} := \mathbf{w}^{(k)} + \eta \mathbf{h}$$

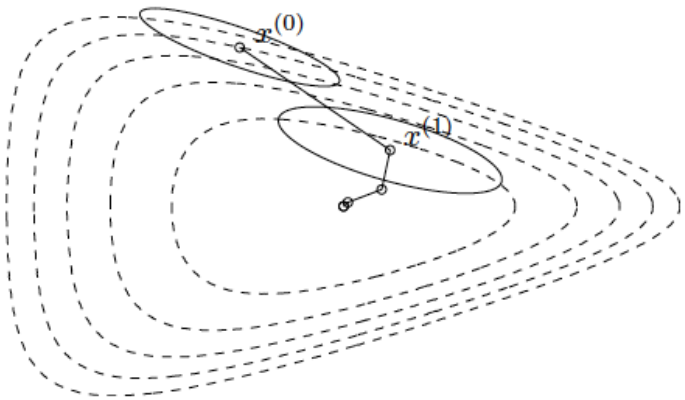
  where η is the learning rate,
- note that it is the sign of $\mathbf{H}(\mathbf{w}^{(k)})$ that indicates whether we will move in the direction of the gradient or in the opposite.
  - if $\mathbf{x}^T \mathbf{H}(\mathbf{w}^{(k)}) \mathbf{x} \geq 0$ (i.e. $\mathbf{H}$ PSD) then $\mathbf{h}$ moves to the oposite of the gradient, to a minimum point.
  - if $-\mathbf{x}^T \mathbf{H}(\mathbf{w}^{(k)}) \mathbf{x} \geq 0$ (i.e. $\mathbf{H}$ NSD) then $\mathbf{h}$ moves to the direction of the gradient, to a maximum point.

Newton's method and the Hessian, [3]

- In Newton's method the optimization direction is $-(\mathbf{H}(\mathbf{w}^{(k)}))^{-1}\nabla_{\mathbf{w}}E(\mathbf{w}^{(k)})$, i.e. we project the gradient on the inverse of the Hessian.
- This is equivalent to a change of coordinates given by $\mathbf{H}(\mathbf{w}^{(k)}))^{1/2}\mathbf{w}$, and thus the definition of a new norm given by: $\mathbf{w}^T(H(\mathbf{w}^{(k)})\mathbf{w}$. Newton does stepest gradient ascent/descent on the new norm.
- Thus we make large steps in the directions that the gradient does not change much (small second derivative) and small steps when the gradient changes a lot (large second derivative).

- Solid lines in the fig. mark the unit ellipsoids centered at $x^{(k)}$ under the Hessian norm, i.e.

$$(x-x^{(k)})^T H(x^{(k)})(x-x^{(k)})$$



Newton's method vs gradient descent, [4]

- Newton gives faster improvement even compared with having the optimal η
- Newton is not applicable when the Hessian is singular
- However doing a matrix inversion $O(d^3)$ at each step can offset the descent advantage.

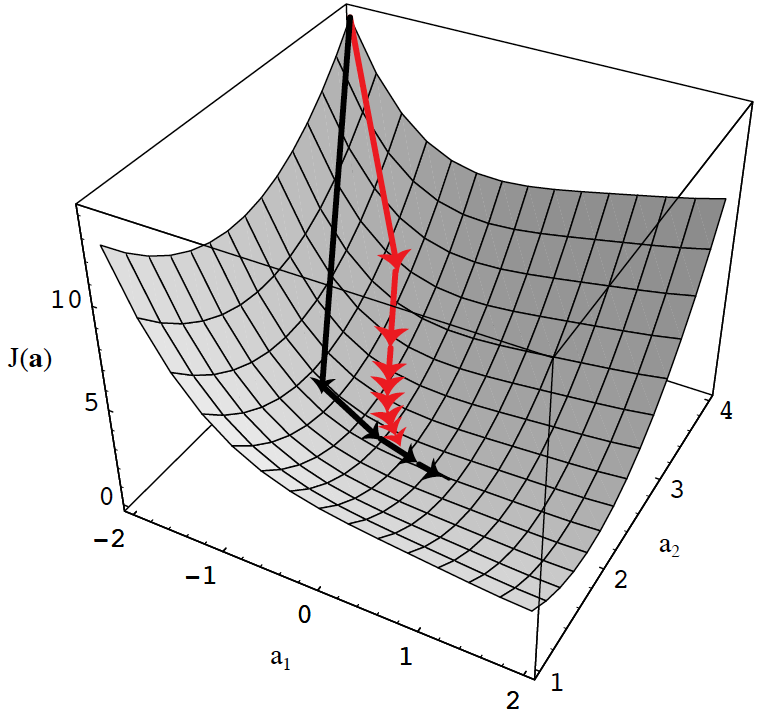
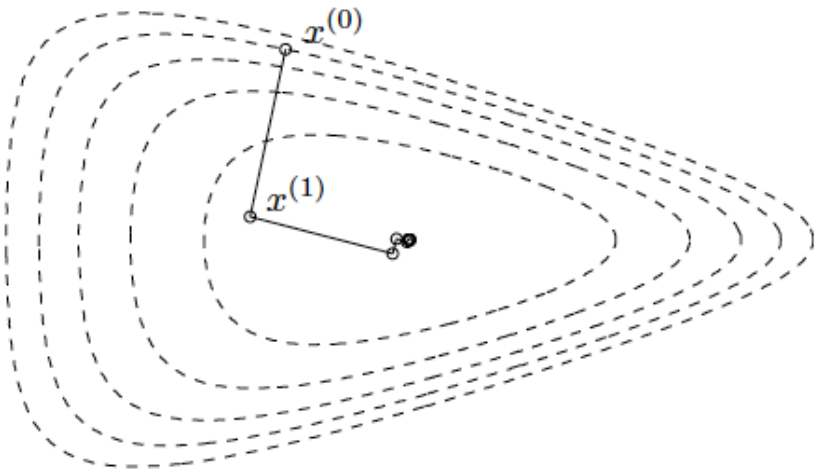
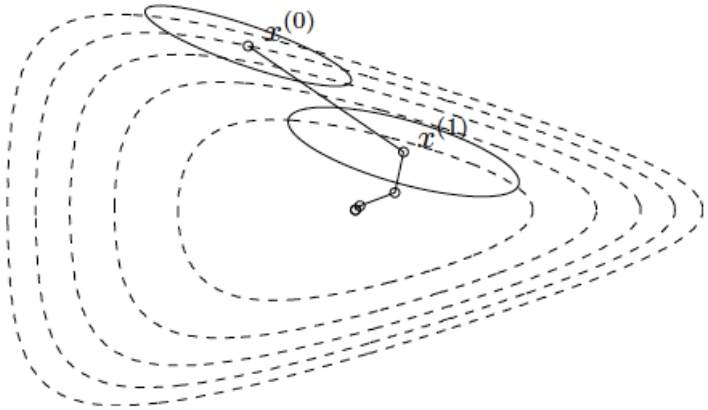


Figure: Sequence of weight vectors given by: gradient descent, black Newton

Newton's method vs gradient descent, [3]





Contour plots of a given function. Left gradient descent steps with exact line search, right Newton's method and backtracking line search.

Newton's method for logistic regression

- Error function:

$$E(\mathbf{w}) = -\sum_{i=1}^{N} y_i \ln \sigma(\mathbf{w}^T\mathbf{x}_i) - \sum_{i=1}^{N}(1-y_i)\ln(1-\sigma(\mathbf{w}^T\mathbf{x}_i))$$

- Gradient:

$$\nabla E(\mathbf{w}) = \sum_{i=1}^{N}(\sigma(\mathbf{w}^T\mathbf{x}_i) - y_i)\mathbf{x}_i = \mathbf{X}^T(\boldsymbol{\sigma} - \mathbf{y})$$

where $\boldsymbol{\sigma} := (\sigma(\mathbf{w}^T\mathbf{x}_1), \ldots, \sigma(\mathbf{w}^T\mathbf{x}_n))^T$

- Hessian:

$$\mathbf{H}(\mathbf{w}) = \nabla\nabla E(\mathbf{w}) = \sum_{i=1}^{N} \sigma_i(1-\sigma_i)\mathbf{x}_i\mathbf{x}_i^T = \mathbf{X}^T\mathbf{R}\mathbf{X}$$

where $\mathbf{R} = diag(\sigma_1(1-\sigma_1), \ldots, \sigma_n(1-\sigma_n))$. Any idea what the quantity $\sigma_i(1-\sigma_i)$?

Newton's method for logistic regression

- Error function:

$$E(\mathbf{w}) = -\sum_{i=1}^{N} y_i \ln \sigma(\mathbf{w}^T\mathbf{x}_i) - \sum_{i=1}^{N}(1-y_i)\ln(1-\sigma(\mathbf{w}^T\mathbf{x}_i))$$

- Gradient:

$$\nabla E(\mathbf{w}) = \sum_{i=1}^{N}(\sigma(\mathbf{w}^T\mathbf{x}_i) - y_i)\mathbf{x}_i = \mathbf{X}^T(\boldsymbol{\sigma} - \mathbf{y})$$

where $\boldsymbol{\sigma} := (\sigma(\mathbf{w}^T\mathbf{x}_1), \ldots, \sigma(\mathbf{w}^T\mathbf{x}_n))^T$

- Hessian:

$$\mathbf{H}(\mathbf{w}) = \nabla\nabla E(\mathbf{w}) = \sum_{i=1}^{N} \sigma_i(1-\sigma_i)\mathbf{x}_i\mathbf{x}_i^T = \mathbf{X}^T\mathbf{R}\mathbf{X}$$

where $\mathbf{R} = diag(\sigma_1(1-\sigma_1), \ldots, \sigma_n(1-\sigma_n))$. Any idea what the quantity $\sigma_i(1-\sigma_i)$? The variance of a Bernouli distribution.

Newton's method for logistic regression

- The update rule is:

$$\begin{aligned}
\mathbf{w}^{(k+1)} &= \mathbf{w}^{(k)} - \mathbf{H}(\mathbf{w}^{(k)})^{-1}\nabla E(\mathbf{w}^{(k)}) \\
&= \mathbf{w}^{(k)} - (\mathbf{X}^T\mathbf{R}\mathbf{X})^{-1}\mathbf{X}^T(\boldsymbol{\sigma} - \mathbf{y}) \\
&= (\mathbf{X}^T\mathbf{R}\mathbf{X})^{-1}[(\mathbf{X}^T\mathbf{R}\mathbf{X})\mathbf{w}^{(k)} - \mathbf{X}^T(\boldsymbol{\sigma} - \mathbf{y})] \\
&= (\mathbf{X}^T\mathbf{R}\mathbf{X})^{-1}\mathbf{X}^T\mathbf{R}[\mathbf{X}\mathbf{w}^{(k)} - \mathbf{R}^{-1}(\boldsymbol{\sigma} - \mathbf{y})]
\end{aligned}$$

- setting $\mathbf{z} := \mathbf{X}\mathbf{w}^{(k)} - \mathbf{R}^{-1}(\boldsymbol{\sigma} - \mathbf{y})$ we can rewrite as:

$$\mathbf{w}^{(k+1)} = (\mathbf{X}^T\mathbf{R}\mathbf{X})^{-1}\mathbf{X}^T\mathbf{R}\mathbf{z}$$

which is known as *iterative reweighted least squares* where at each step k we recompute the $\mathbf{R}$ weighting matrix.

Least squares/reweighted least squares

- Standard least squares, optimization problem

$$\min_{\mathbf{w}} \sum_{i}^{N} (y_i - \mathbf{w}^T \mathbf{x}_i)^2 = \min_{\mathbf{w}} (\mathbf{y} - \mathbf{X}\mathbf{w})^T (\mathbf{y} - \mathbf{X}\mathbf{w})$$

- Standard least squares, solution:

$$\mathbf{w} = (\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T \mathbf{y}$$

- reweighted least squares, optimization problem

$$\min_{\mathbf{w}} \sum_{i}^{N} R_i (y_i - \mathbf{w}^T \mathbf{x}_i)^2 = \min_{\mathbf{w}} (\mathbf{y} - \mathbf{X}\mathbf{w})^T \mathbf{R} (\mathbf{y} - \mathbf{X}\mathbf{w})$$

- reweighted least squares, solution:

$$\mathbf{w} = (\mathbf{X}^T \mathbf{R} \mathbf{X})^{-1} \mathbf{X}^T \mathbf{R} \mathbf{y}$$

the contribution of the ith instance is weighted by R_i which as we saw was the variance of its prediction.

On the Hessian/cost function of the logistic regression

$$\begin{aligned}\mathbf{H}(\mathbf{w}) &= \nabla\nabla E(\mathbf{w}) = \sum_{i=1}^{N} \sigma_i(1-\sigma_i)\mathbf{x}_i\mathbf{x}_i^T \\ &= \mathbf{X}^T\mathbf{R}\mathbf{X}\end{aligned}$$

- The Hessian is a function of $\mathbf{w}$ and *not a constant*
- $\forall\ \mathbf{z}, \mathbf{z}^T\mathbf{H}\mathbf{z} > 0$ thus $\mathbf{H}$ is positive definite (except when $||\mathbf{w}|| \to \infty$, in which case $\mathbf{H} \to 0$, comments bellow).
- and $E(\mathbf{w})$ is a convex function with a unique minimum.

Logistic regression cost function drawbacks [5, 2] I

- It uses the Maximum Likelihood principle
- as all ML methods it is prone to overfitting
- In particular if the problem is linearly seperable
  - then the negative likelihood is minimized by finding the direction $\mathbf{w}$ for which $\sigma(\mathbf{w}^\mathsf{T}\mathbf{x}) = 1/2$ i.e. $\mathbf{w}^\mathsf{T}\mathbf{x} = 0$ and sending $||\mathbf{w}|| \to \infty$
  - then $P(C_1|\mathbf{x}) = 1, \forall \mathbf{x} \in C_1)$ and $P(C_2|\mathbf{x}) = 0, \forall \mathbf{x} \in C_2)$
  - and the sigmoid function converges to a Heavyside function in the feature space.
  - we also have: $\sigma_i(1 - \sigma_i) \to 0$ thus $\mathbf{H} \to 0$.
- this assigns maximum probability mass in the training samples but does not generalise,
- and there is a continium of such equivalent solutions which we cannot distinguish

Logistic regression cost function drawbacks [5, 2] II

- Solution:
  - Include a prior on the model parameters, e.g. Gaussian/Laplacian, thus going the Maximum A Posteriori way
  - or equivalently use a regularization terms such as $||\mathbf{w}||_2^2$ or $||\mathbf{w}||_1$

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